

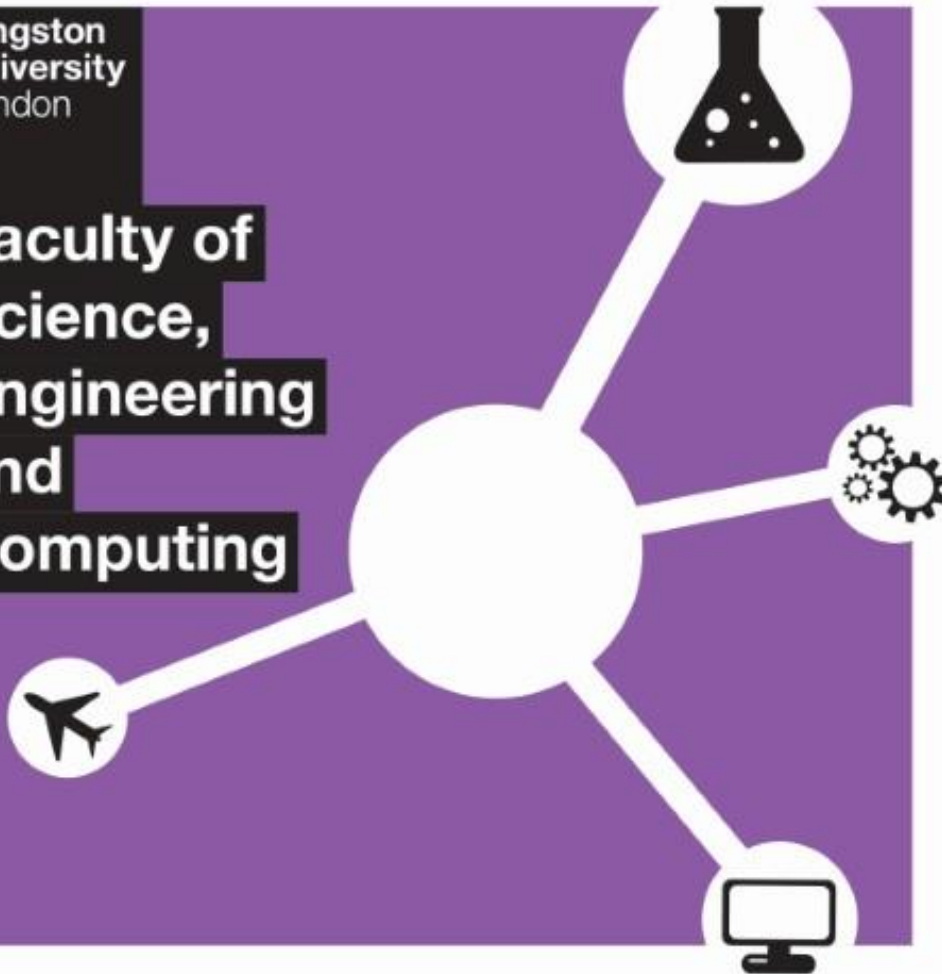
Class 1 – TTC - Props

- **None**

House Keeping

Kingston
University
London

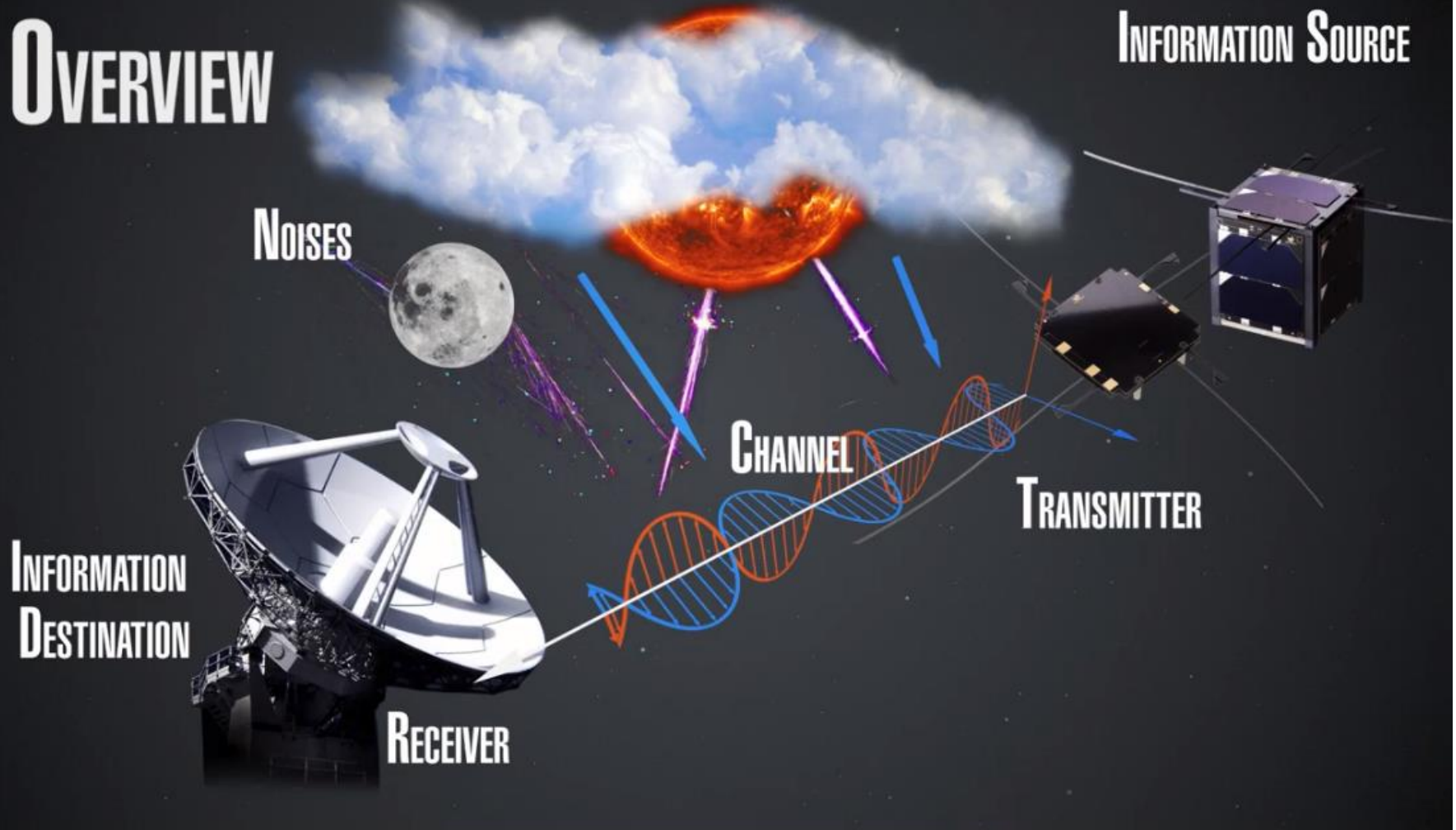
Faculty of
Science,
Engineering
and
Computing



KEEP
CALM
AND
TURN PHONE
TO SILENT MODE

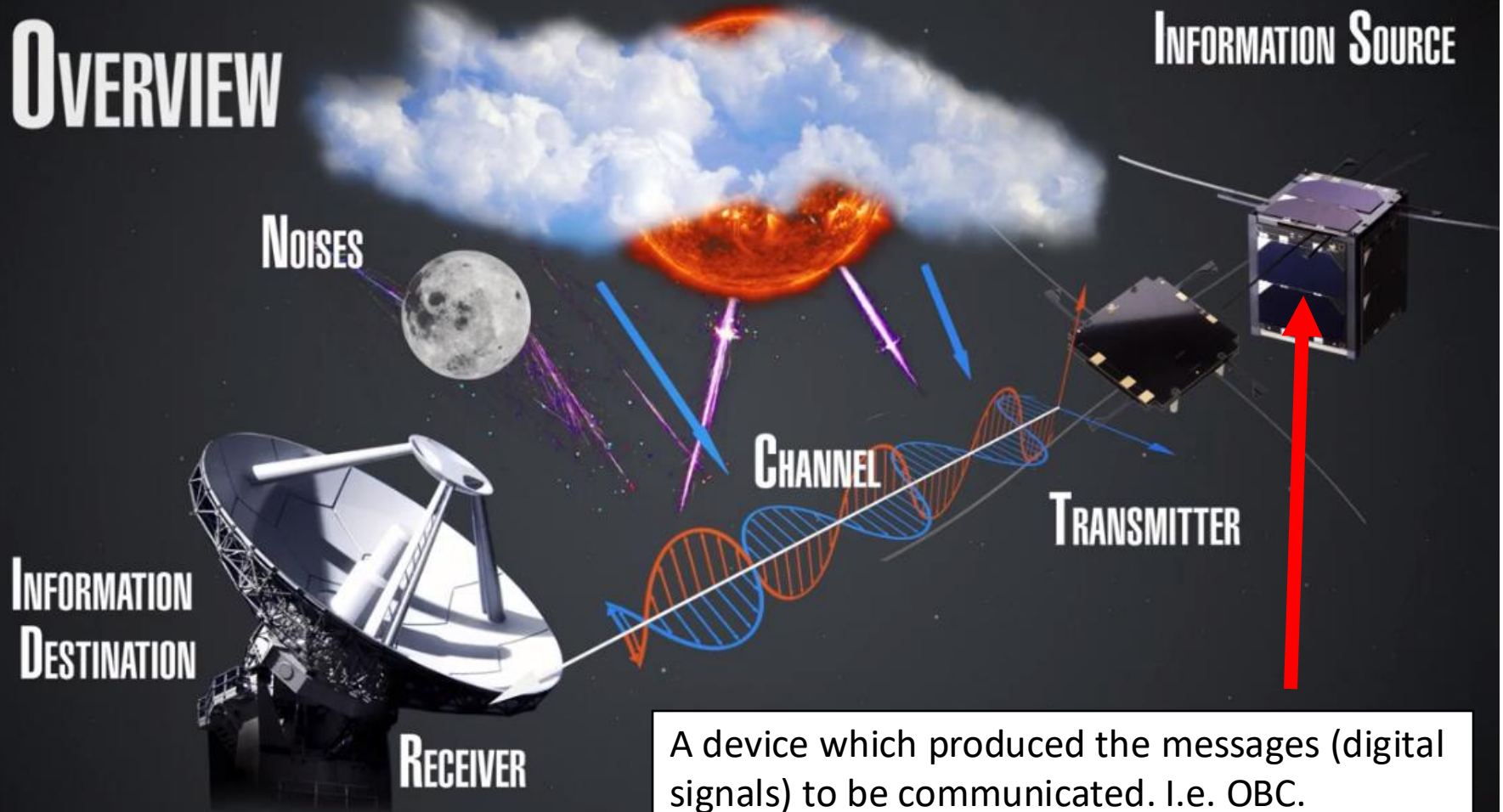
Introduction

OVERVIEW



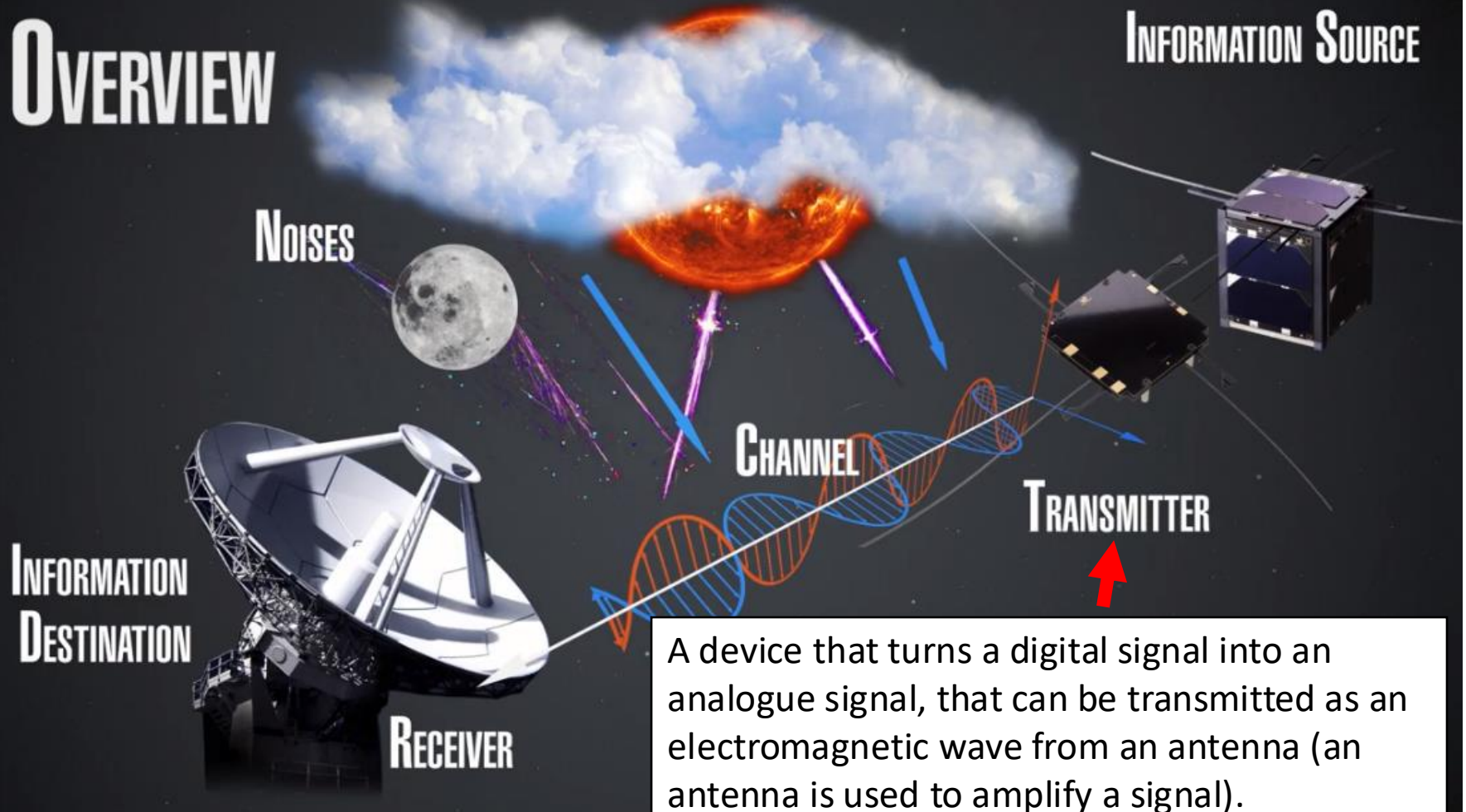
Introduction

OVERVIEW



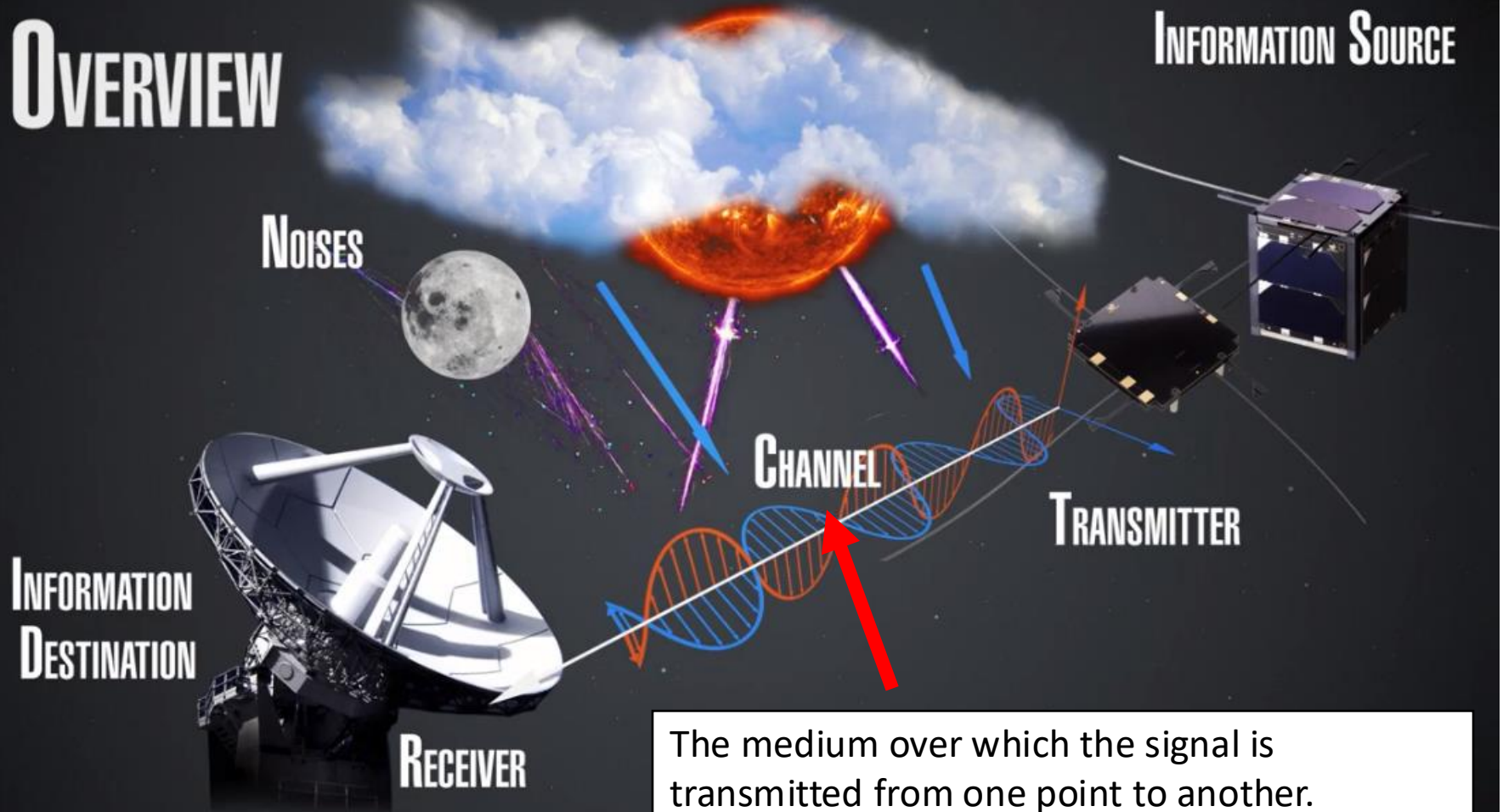
Introduction

OVERVIEW



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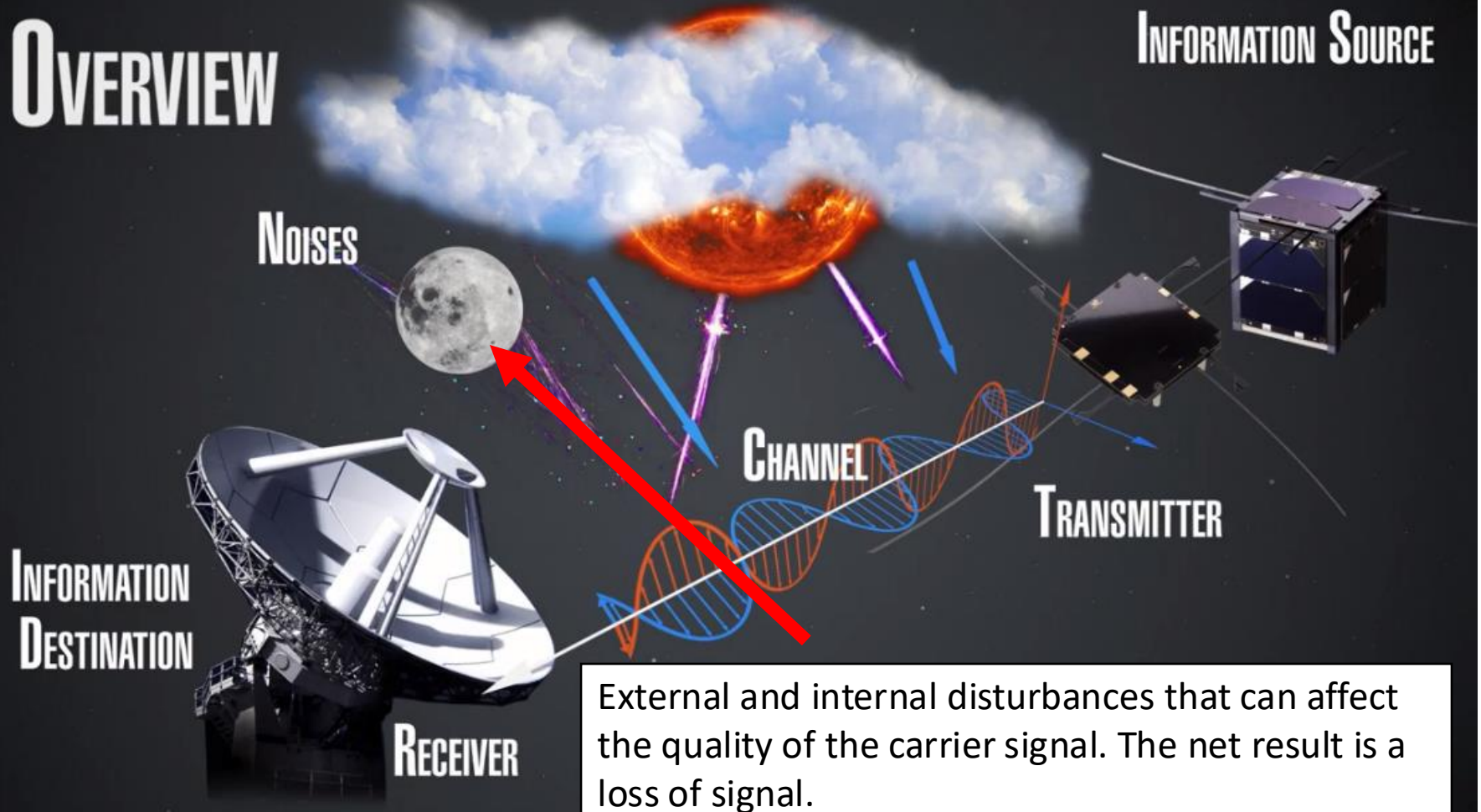
OVERVIEW



The medium over which the signal is transmitted from one point to another.

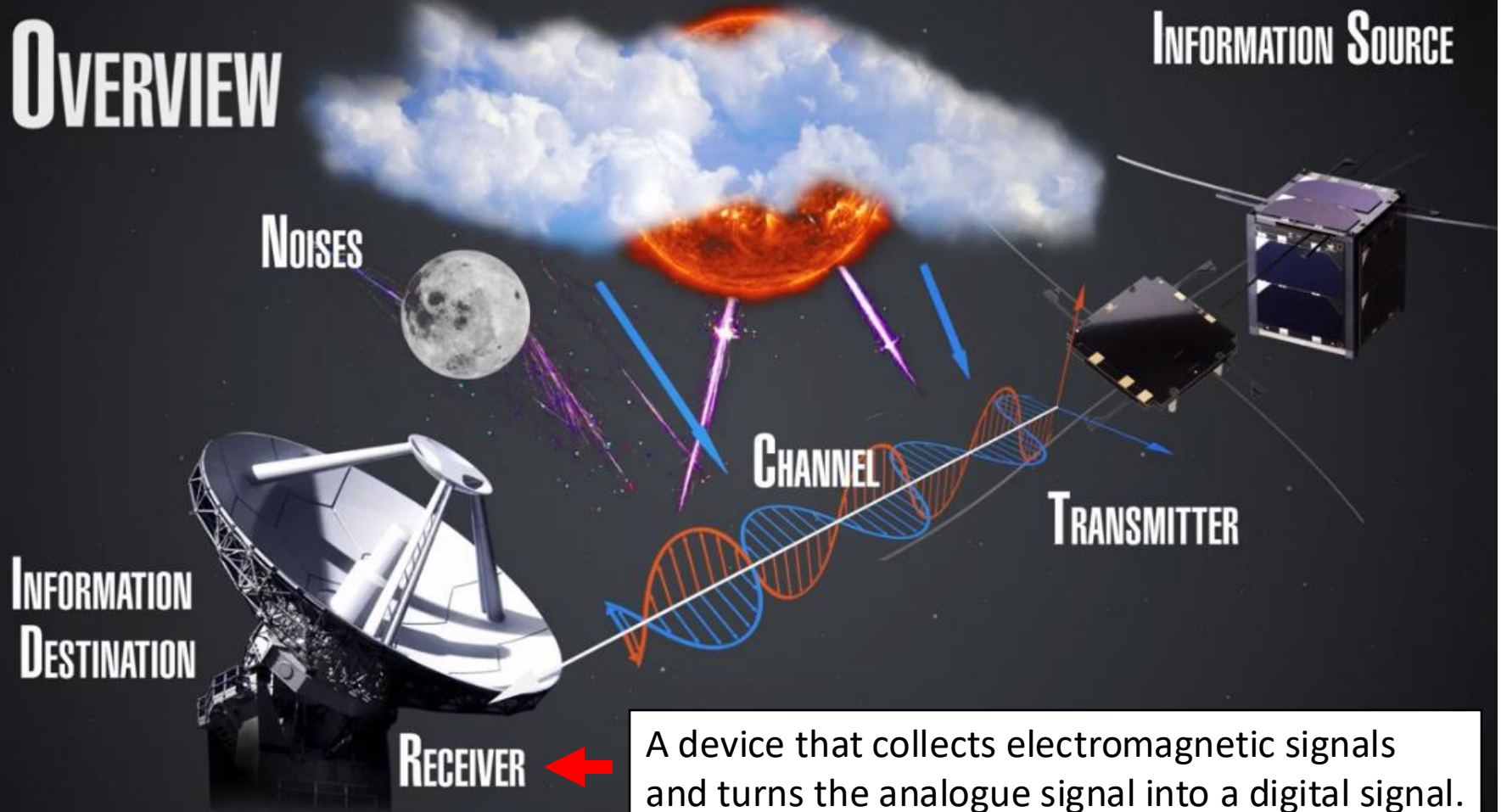
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OVERVIEW



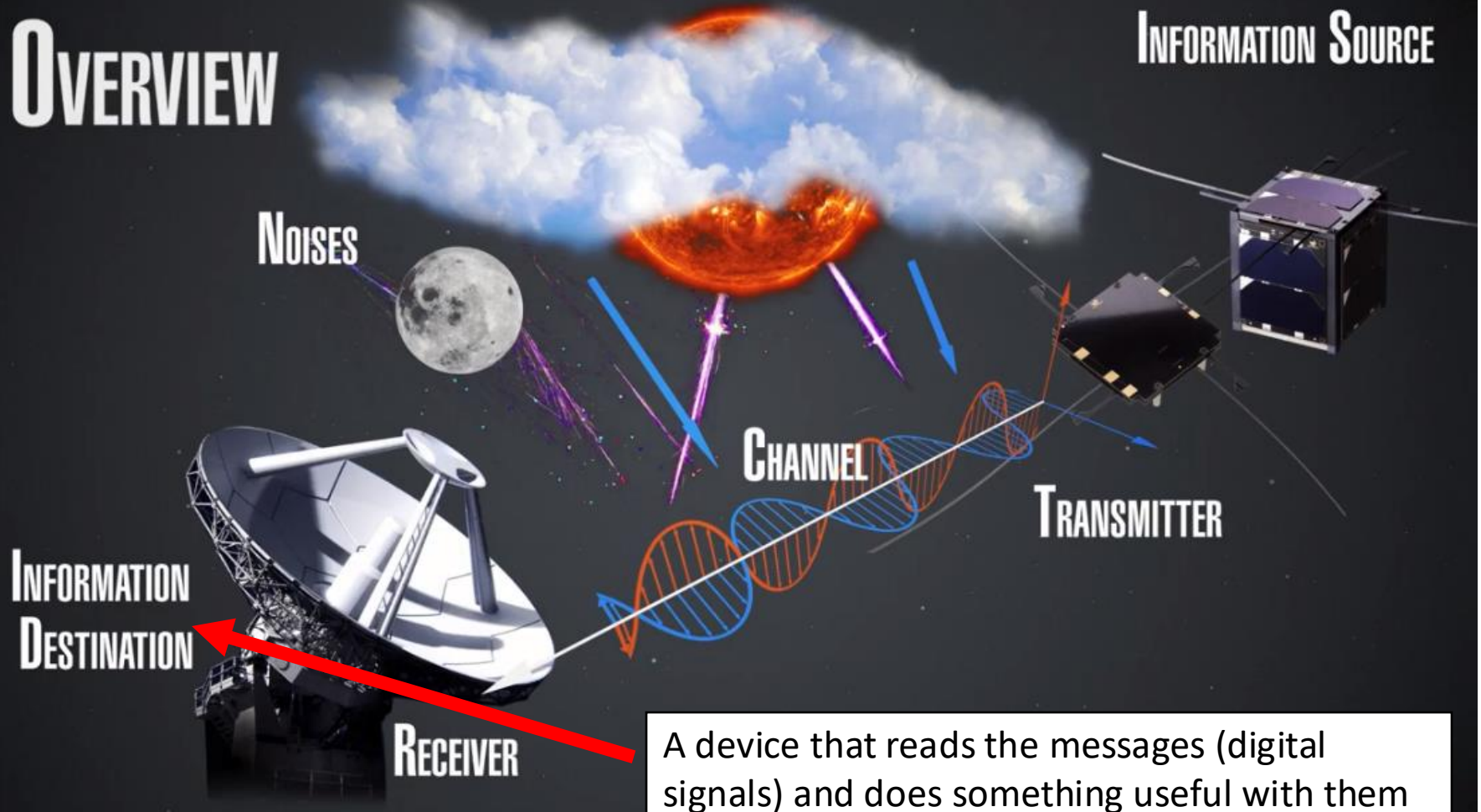
Introduction

OVERVIEW



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OVERVIEW



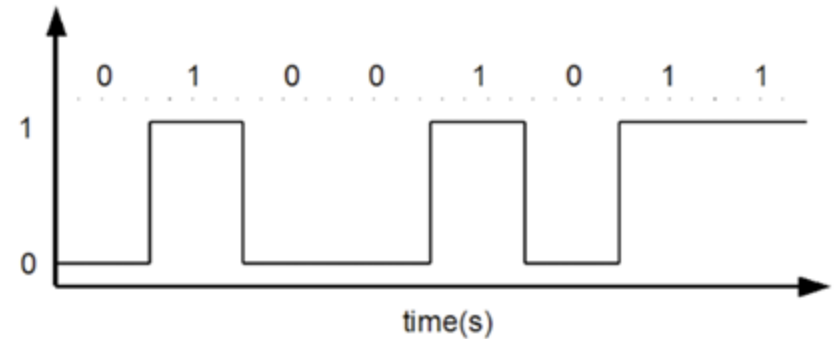
Introduction - Waves

Continuous Vs Discrete Signals

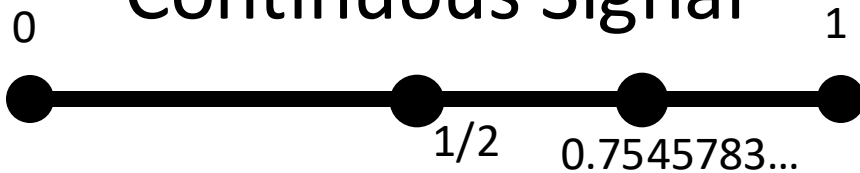
Discrete Signal



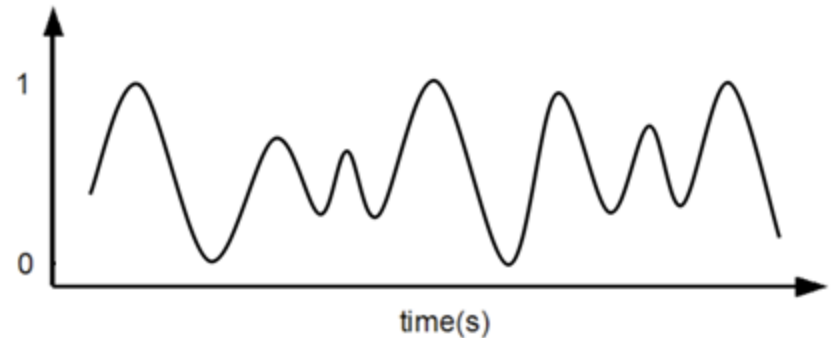
Only a defined 'n' number of solutions, in this case 0 and 1. Typical in digital signals that can be understood electronically.



Continuous Signal

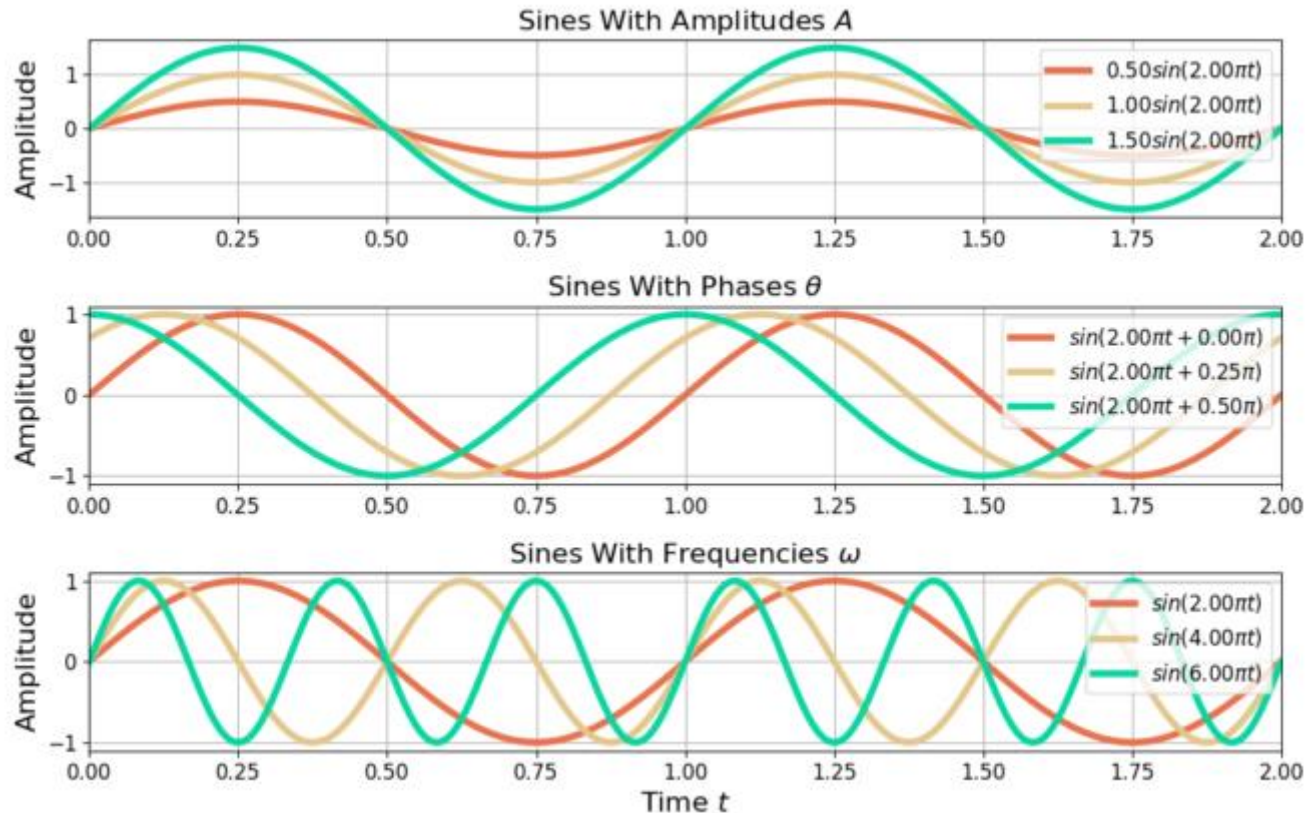


Infinite number of solutions between 0 and 1. Typical in analog signals that can be used to transmit and receive electromagnetic signals.



Introduction - Waves

Sine waves have three parameters – **Amplitude**, **Phase** and **Frequency**



Amplitude represents the magnitude, or size, of a wave.

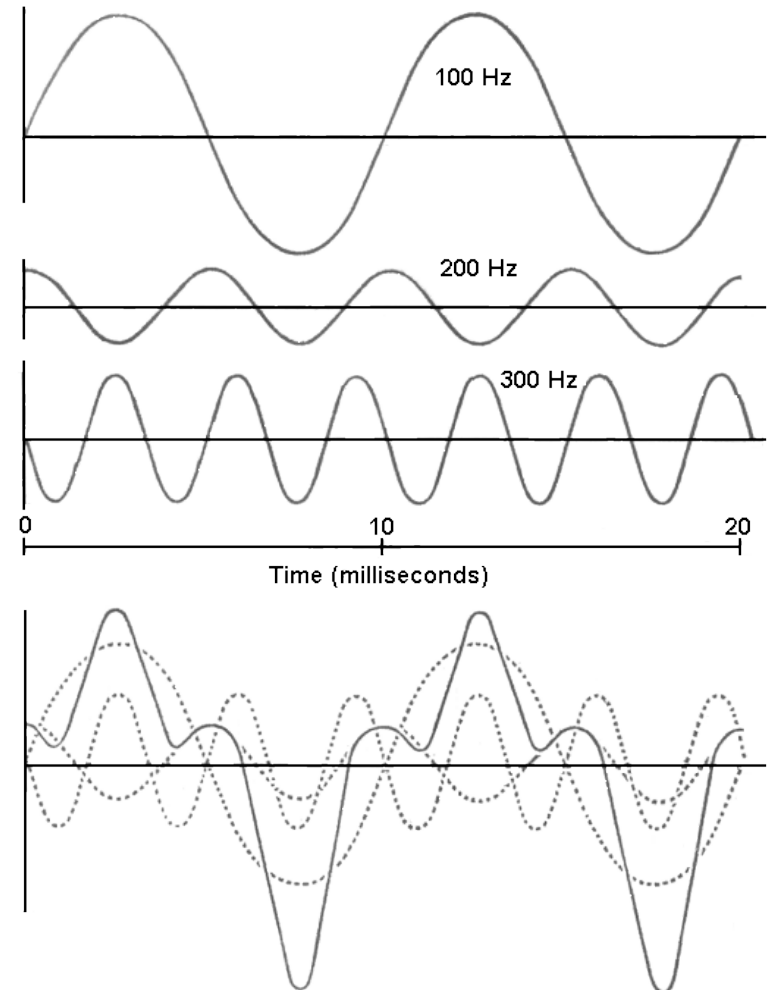
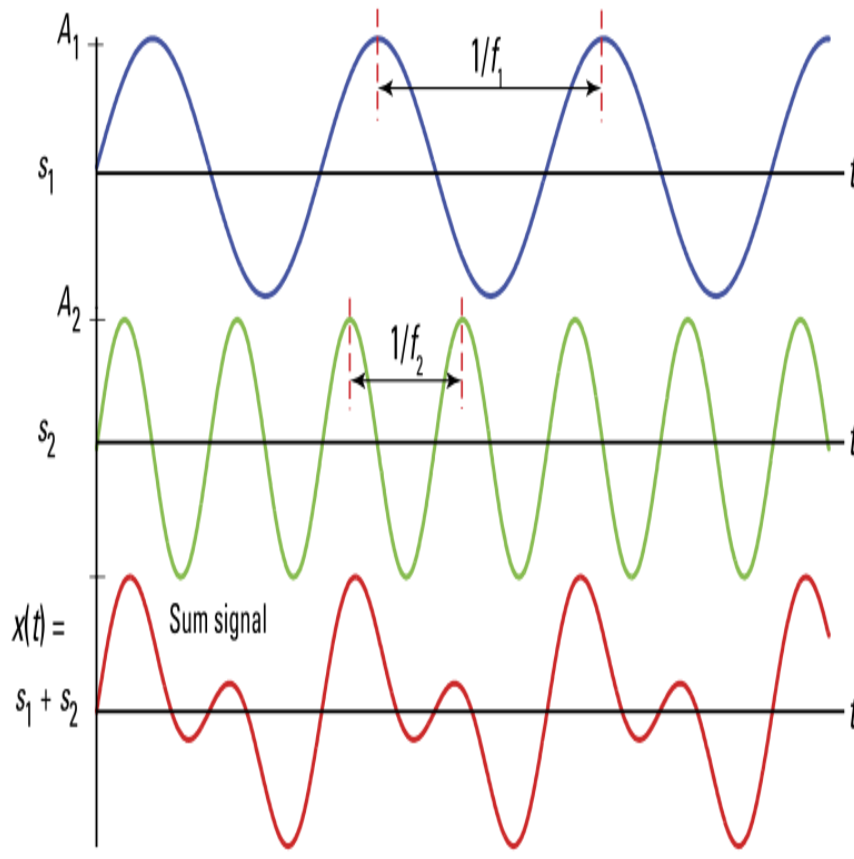
If two waves with the **same frequency** are **perfectly aligned**, they are said to have a phase difference of zero and are **“in phase.”** Sine waves with the **same frequency but in a different cycle** are **“out of phase.”**

Frequency measures the number of wave cycles that occur in one second. Note angular frequency is related to frequency by $\omega = 2\pi f$. Also, frequency is related to time-period (T) $f = \frac{1}{T}$

So...
$$y(t) = A \sin\left(\frac{2\pi nt}{T} + \theta\right)$$
 $n = \text{harmonic number (integer)}$

Introduction - Waves

Addition of sine waves...



Sine Wave [Calculator](#) So, waves in proximity of each other will interfere with each other!

Introduction - Waves

In fact, using Fourier transform functions we can describe any repeating (i.e. periodic) wave function as an addition of sin and cosine elements....

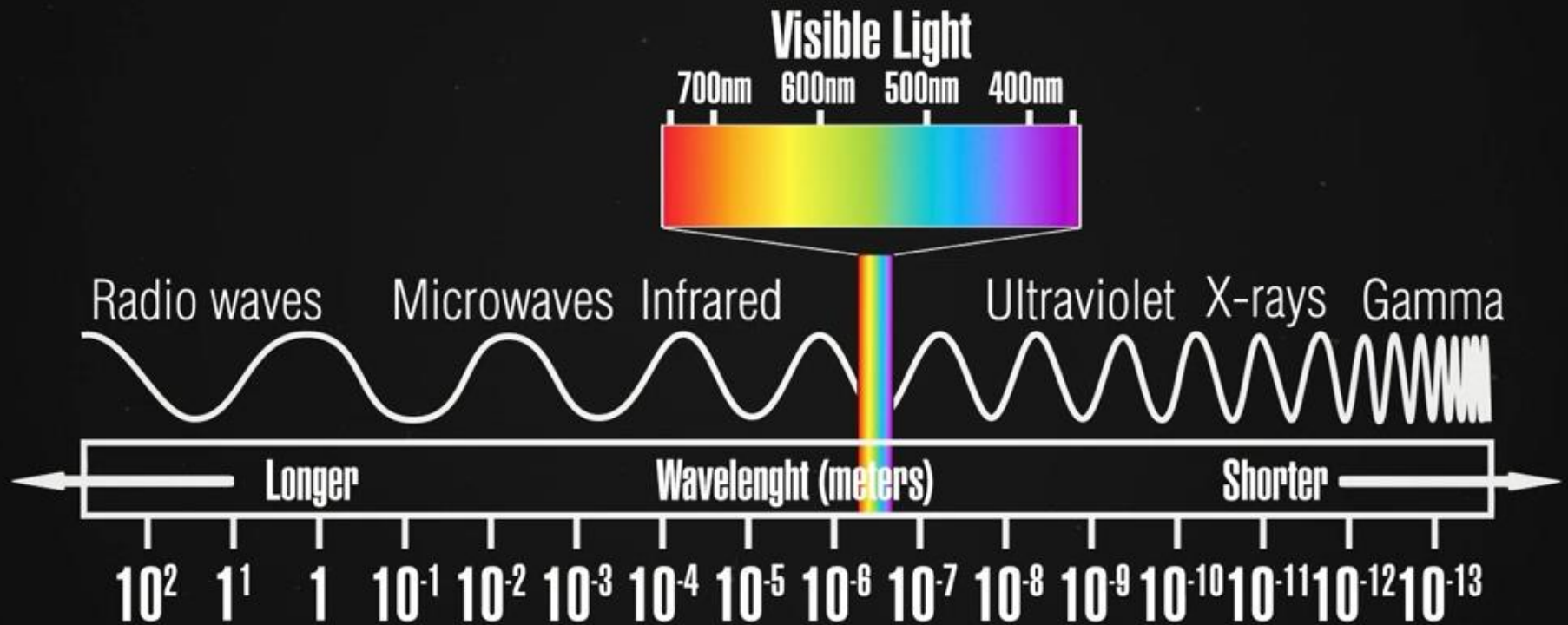
$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2\pi n t}{T}\right) + b_n \sin\left(\frac{2\pi n t}{T}\right)$$

Where the coefficients are given as...

$$a_0 = \frac{2}{T} \int_0^T f(t) dt \quad a_n = \frac{2}{T} \int_0^T f(t) \cos\left(\frac{2\pi n t}{T}\right) dt \quad b_n = \frac{2}{T} \int_0^T f(t) \sin\left(\frac{2\pi n t}{T}\right) dt$$

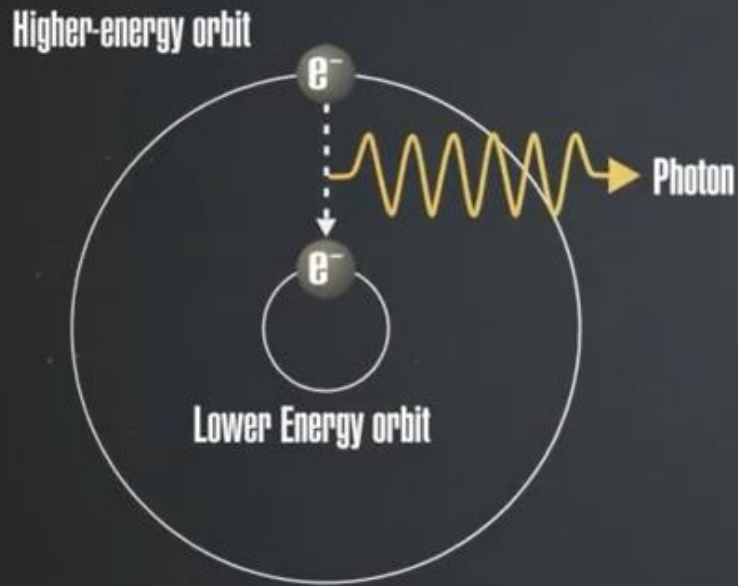
Note – Due to $f(t)$ appearing on both sides of the equation, it is common to use numerical analysis to understand and compute the coefficients.

Introduction – Electromagnetic Waves

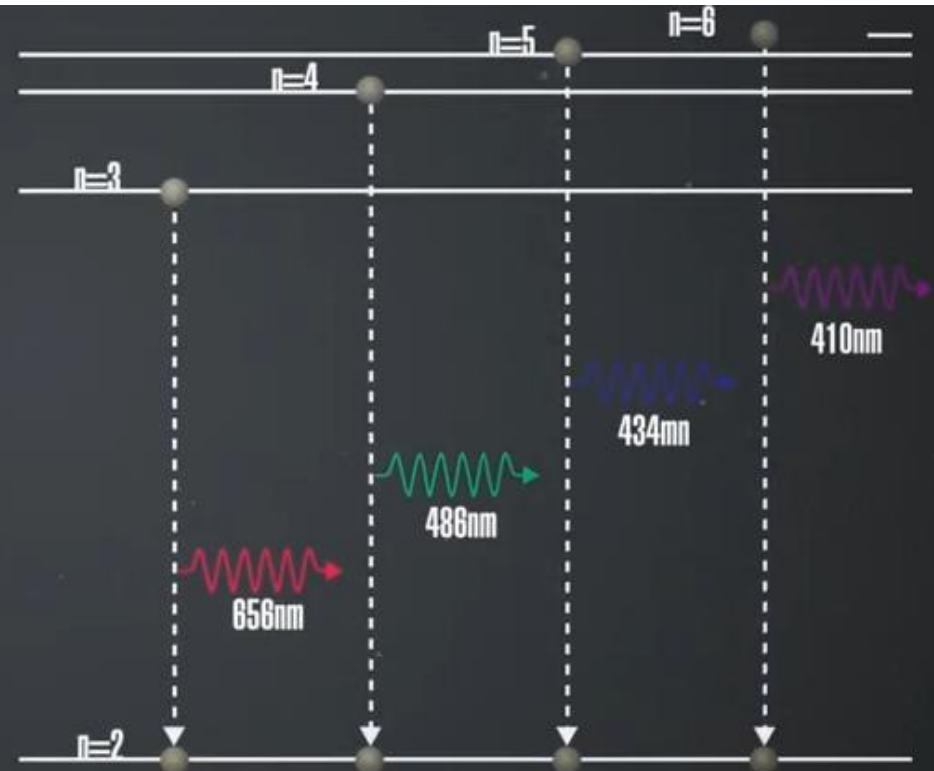


Introduction – Electromagnetic Waves

Where do electromagnetic waves come from?



(a) Electronic emission transition



(b) Balmer series transitions

Introduction – Electromagnetic Waves

Electromagnetic Wave

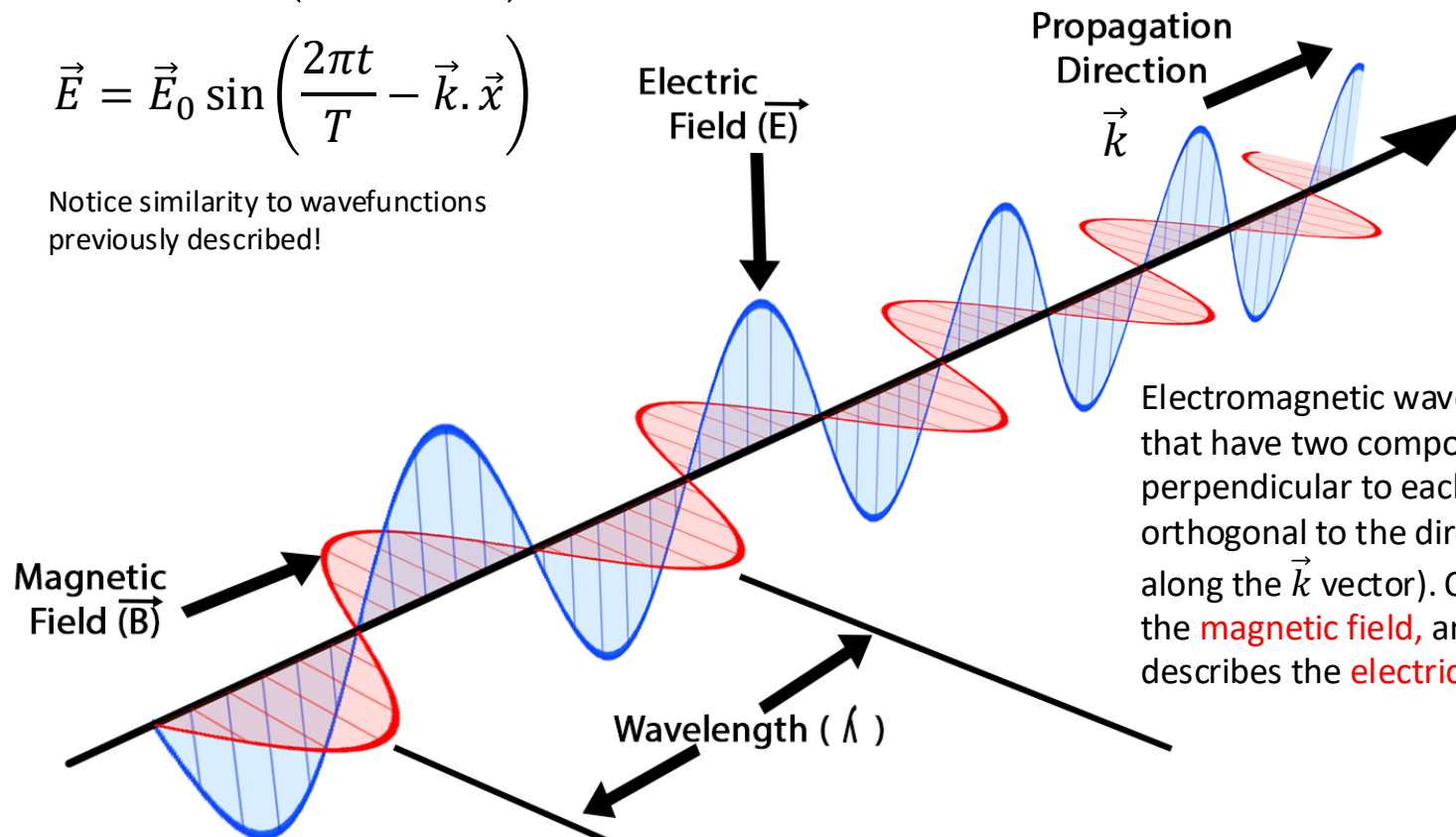
$$\vec{B} = \vec{B}_0 \sin\left(\frac{2\pi t}{T} - \vec{k} \cdot \vec{x}\right)$$

$$\vec{E} = \vec{E}_0 \sin\left(\frac{2\pi t}{T} - \vec{k} \cdot \vec{x}\right)$$

Notice similarity to wavefunctions previously described!

$$c = 1/\sqrt{\mu_0 \epsilon_0}$$

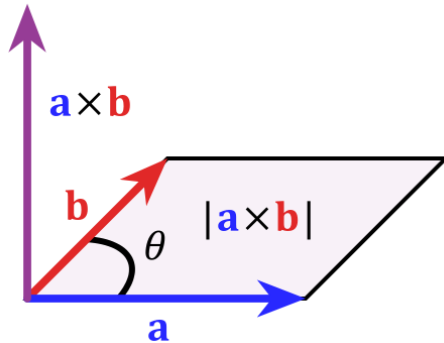
μ_0 = vacuum magnetic permeability
 ϵ_0 = permittivity of free space



Electromagnetic waves are **transverse waves** that have two components that are perpendicular to each other and oscillate orthogonal to the direction of motion (i.e. along the $\vec{k}$ vector). One component describes the **magnetic field**, and the other component describes the **electric field**.

Introduction – Electromagnetic Waves

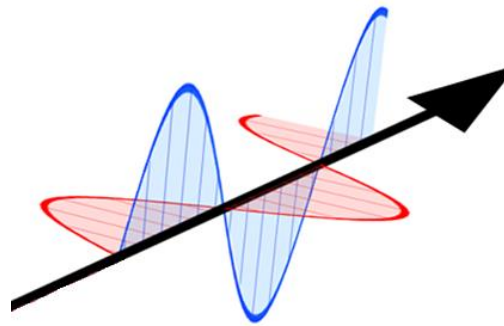
Reminder (cross product)



The **Poynting vector** is the **Power flux** of the electromagnetic field.

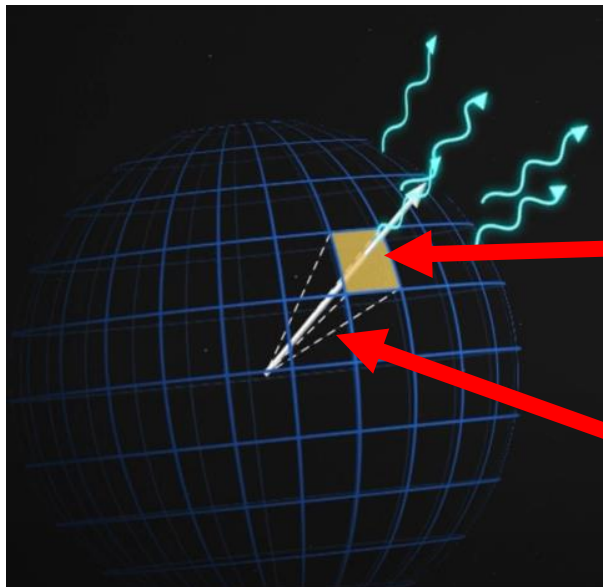
$$\vec{P} = \frac{1}{\mu_0} \vec{B} \times \vec{E}$$

The free wave carries energy in the direction of its propagation since $\vec{k}$ and $\vec{P}$ are colinear.



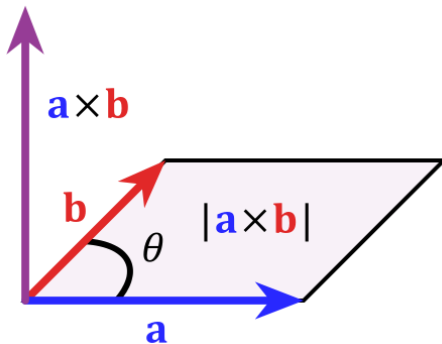
Power flux – The **energy per unit time passing through a unit area** (leaving or entering the surface)

Energy Density – The **energy per unit volume inside the surface**



Introduction – Electromagnetic Waves

Reminder (cross product)



Total Power per unit area (for plane waves)

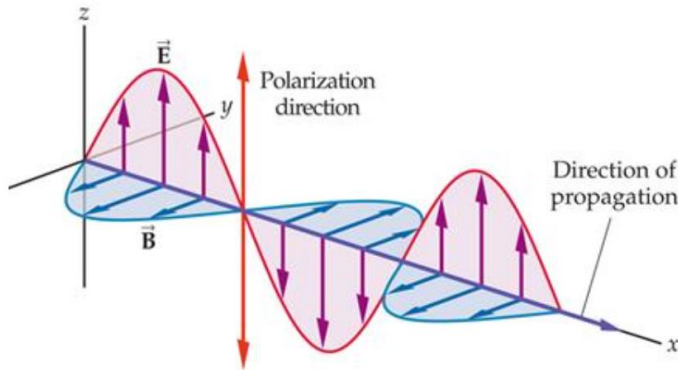
$$|\vec{P}| = \frac{1}{\mu_0} |\vec{B} \times \vec{E}| = c\epsilon_0 |\vec{E}|^2 = \frac{c}{\mu_0} |\vec{B}|^2$$

Average Power (Intensity) of a free electromagnetic wave per unit area (for plane waves)

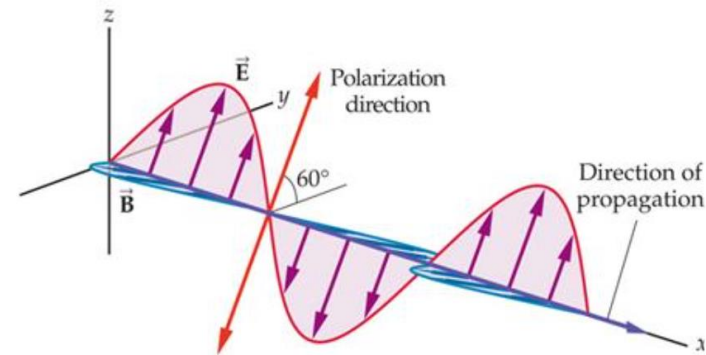
$$\langle |\vec{P}| \rangle = \frac{1}{2} c\epsilon_0 |\vec{E}|^2 = \frac{c}{2\mu_0} |\vec{B}|^2$$

Introduction – Electromagnetic Waves

Polarisation

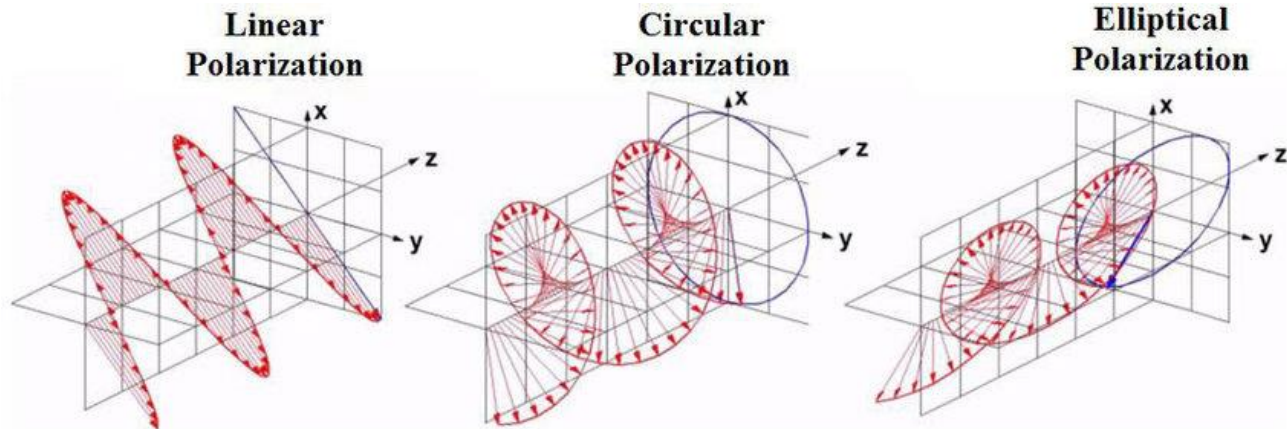


This wave is polarized in z -direction



This wave is polarized in a direction at an angle of 60° with y -axis

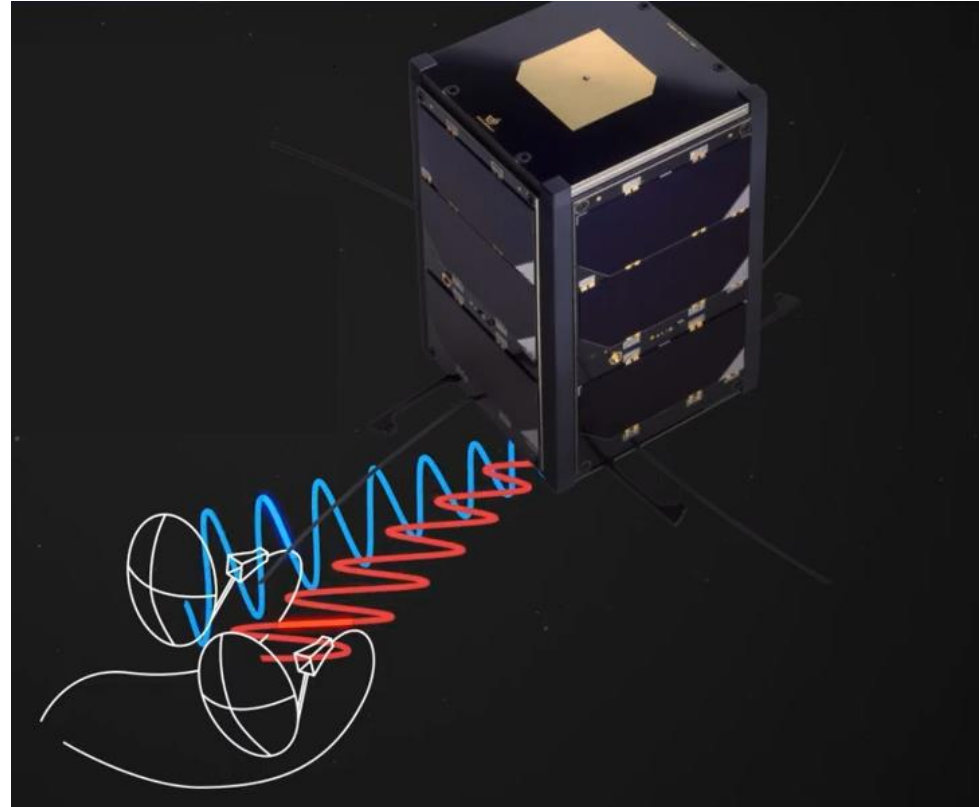
But what if the polarisation angle continually rotated?



Introduction – Electromagnetic Waves

Polarisation is the direction of oscillation with respect to some reference direction

- Almost all EM waves will interfere with each other (e.g. the electric or magnetic field components in the sine waves will add up and distort the signal resulting in the loss of information).
- However, in a special case, EM waves that are orthogonal to each other (e.g. the electric and magnetic components of the field of two separate waves are at 90 degrees to each other) will not contribute to each other's electrical or magnetic fields and hence do not distort or interfere with each other.
- In satellites we use this to be able to receive and transmit without distorting signals in close proximity.
- In satellites we also use circular polarisation as it makes the orientation of the antennas agnostic to the orientation of the electromagnetic waves.

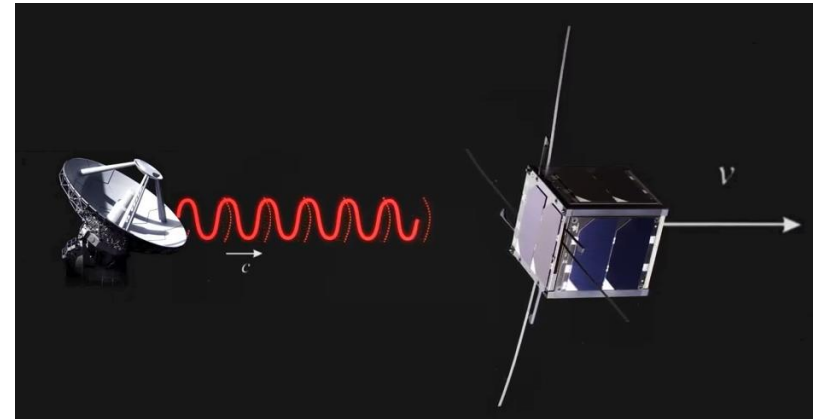


Introduction – Electromagnetic Waves

Doppler Effect

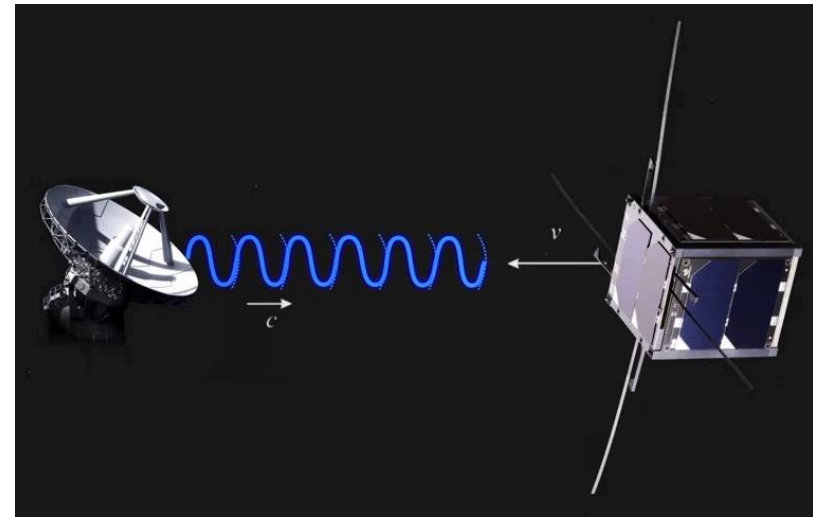
Receding from source...

$$\frac{f_O}{f_E} = \sqrt{\frac{1 - \frac{v}{c}}{1 + \frac{v}{c}}} \quad \text{When } v \ll c \text{ then } \frac{f_O}{f_E} \cong 1 - \frac{v}{c}$$



Approaching the source...

$$\frac{f_O}{f_E} = \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}} \quad \text{When } v \ll c \text{ then } \frac{f_O}{f_E} \cong 1 + \frac{v}{c}$$



f_O = frequency of observed wave

f_E = frequency of emitted wave

v = velocity of the observer with respect to the source

Introduction – Electromagnetic Waves

Doppler Effect – An example LEO Satellite

Given a satellite moving at $v = 7.5$ km/s, receiving an S-Band signal emitted from a ground station with a $f_E = 2.4$ GHz, what is the doppler shift before and after the pass and what is the total change in frequency?



$$\frac{f_O}{f_E} \cong 1 \pm \frac{v}{c} = 1 \pm \frac{7.5}{30000} = 1 \pm 0.000025$$

$$f_O = f_E(1 \pm 0.000025)$$

$$f_O = 2.4(1 \pm 0.000025)$$

$$f_{Omin} = 2.39994 \text{ GHz} \quad f_{Omax} = 2.40006 \text{ GHz}$$

$\Delta f = 120$ kHz – Although small compared to the emitted frequency, this is significant when considering modulation of a carrier wave (see later section).

Introduction – Communication Bands

IEEE Classification – Radio wave frequency ranges and Band designations

Band Designation	Meaning	Frequency Range
HF	High Frequency	0.003 to 0.03 GHz
VHF	Very High Frequency	0.03 to 0.3 GHz
UHF	Ultra High Frequency	0.3 to 1 GHz
L	Long wave	1 to 2 GHz
S	Short wave	2 to 4 GHz
C	Between S and X	4 to 8 GHz
X	X for cross	8 to 12 GHz
Ku	Kurz-under	12 to 18 GHz
K	Kurz (German for short)	18 to 27 GHz
Ka	Kurz-above	27 to 40 GHz
V	-	40 to 75 GHz
W	-	75 to 110 GHz
mm or G	Milimeter	110 to 300 GHz

Introduction – Communication Bands

IEEE Classification – Radio wave frequency ranges and Band designations

- HF – Shortwave radio, radiotelephony
- VHF – TV broadcasting, FM radio
- UHF – TV broadcasting, cellphones, GNSS (Global Navigation Satellite System) signals
- L-band – GPS carriers and satellite mobile phones (such as Iridium)
- S-band – Weather radar, surface radar and some communication satellites (especially NASA communication with the ISS)

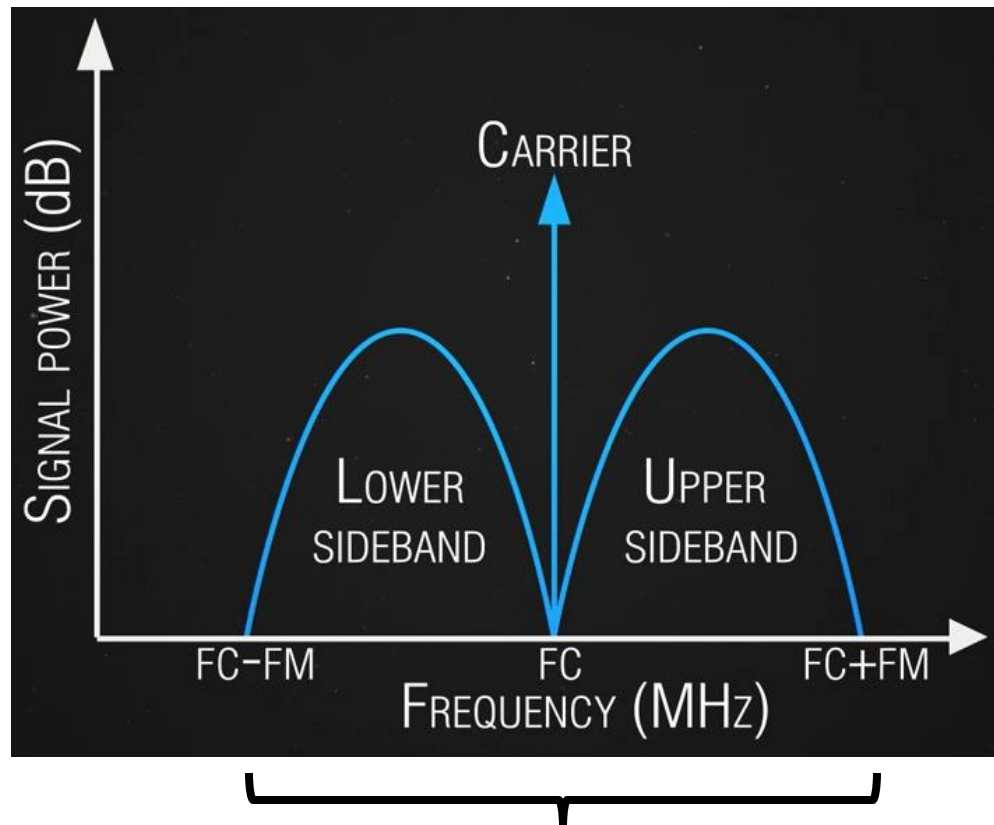
Introduction – Communication Bands

IEEE Classification – Radio wave frequency ranges and Band designations

- C-band – Primarily used for satellite communication and satellite TV
- X-band – Used in radar applications (synthetic aperture and phased array radars), weather monitoring and air traffic control
- Ku-band – Used for satellite communications in Europe
- Ka-band – Communications satellites

Introduction – Communication Bands

Bandwidth and Carrier

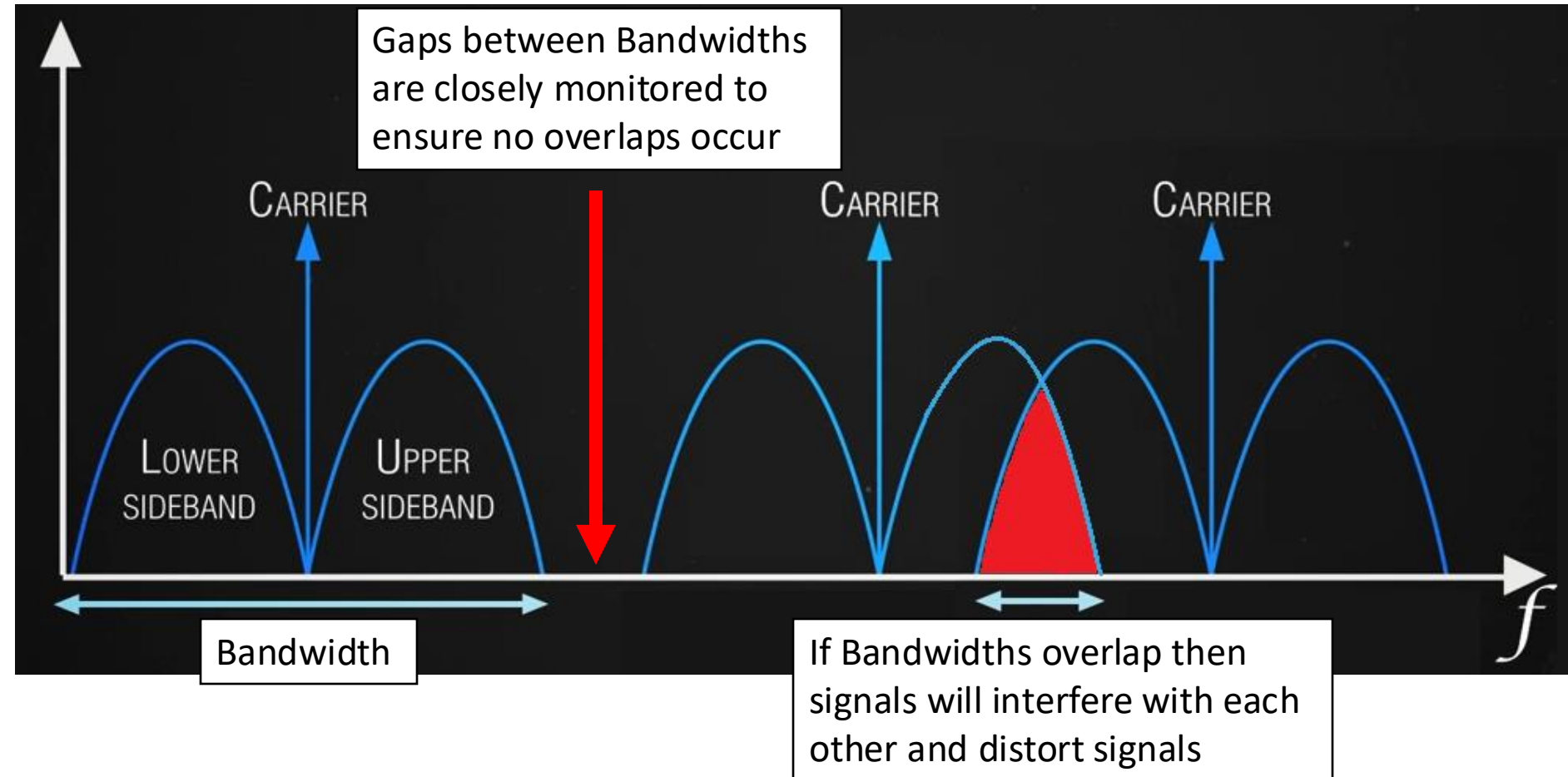


Bandwidth is the frequency range between a minimum and maximum

- Communication Bands are set by the International Telegraph Union, can be split up into smaller frequency ranges.
- You can apply to the ITU to register a communication frequency for satellite communication.
- A frequency will be given within a tight bandwidth (to minimise interference)
- If a satellite is not launched and transmitting within a certain time frame, the ITU will retake granted frequency so it can be reallocated.
- Hardware is made specific to a frequency, so losing a frequency in this matter can cost £M's.

Introduction – Communication Bands

Bandwidth and Carrier



Emitting and Receiving Electromagnetic Waves



Converting Data into Digital Signals

Digitalization - CONVERTING A CONTINUOUS SIGNAL INTO A DISCRETE SEQUENCE!



The analog signal is sampled at regular intervals

Each sample is then converted into a digital value (e.g., 0 or 1)

Introduction

- A satellite is a tele-operated system
 - that can be reached via radio links
 - ...only few minutes per orbits
 - ...every 90 min or more
- Operators need to be able:
 - To monitor the health of the spacecraft : **Telemetry**
 - To determine its position
 - To influence its behaviour : **Command**
- System composed of Space and Ground segments

Full YouTube Course can be found [here](#)

Introduction

Space Segment:

- Collects data from subsystems and relays them to the ground
- Receives telecommands from the ground and dispatches them to the subsystems
- Relays ranging tones

Ground Segment:

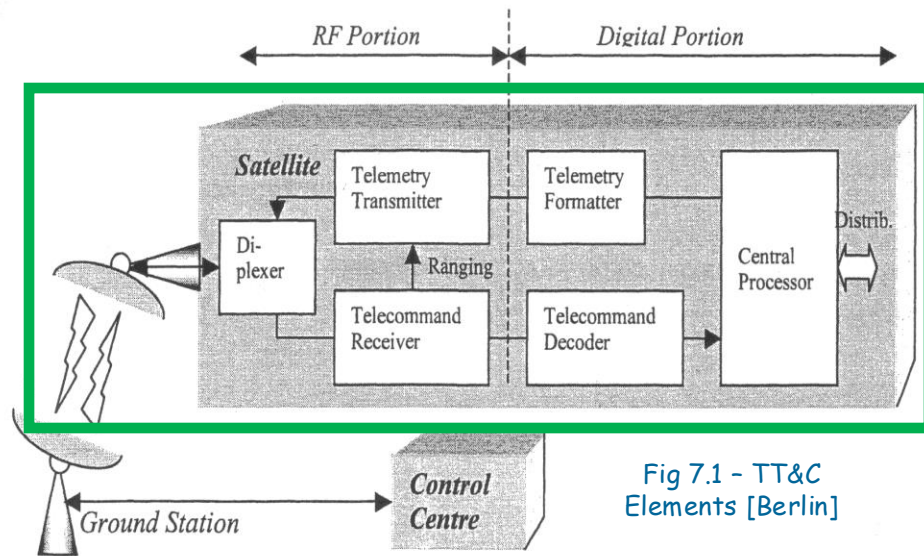
- Mirrors the spacecraft TT&C functions (...just an antenna and some HW)
- Has a control centre where the craft are operated, telemetry processed and commands prepared.

The RF portion:

- Deals with the transmission of the data to and from the digital portion and the Ground Segment

The digital portion:

- Collects and converts telemetry data
- Routes the telecommands to the subsystems



TT&C made up of a Digital and a Radio Frequency (RF) portion.

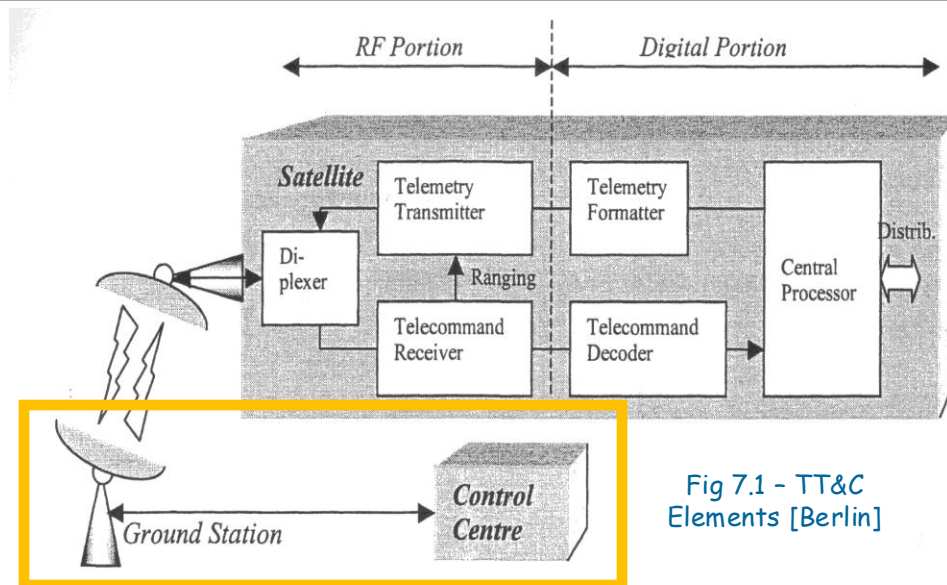
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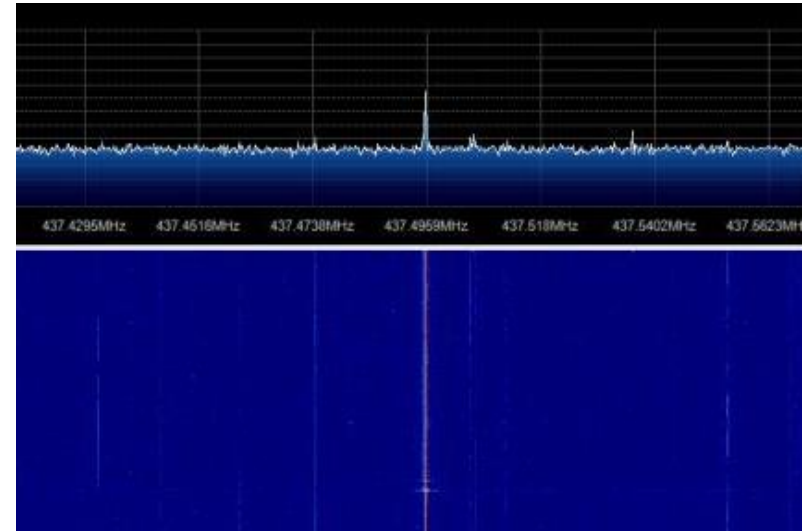
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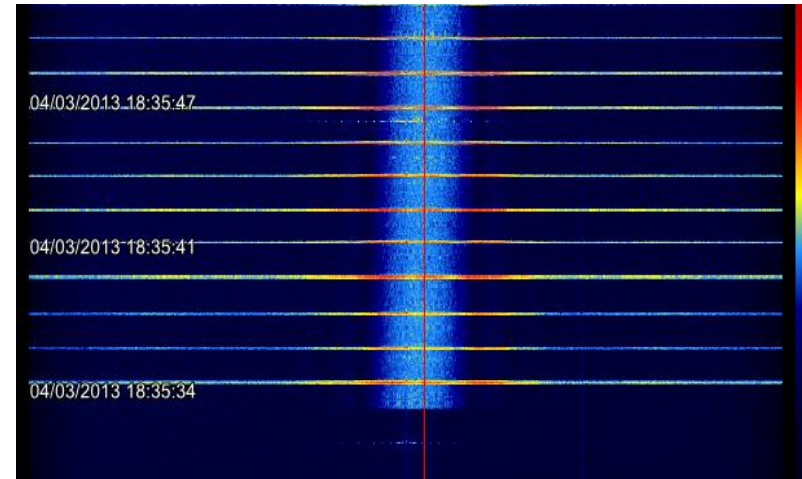
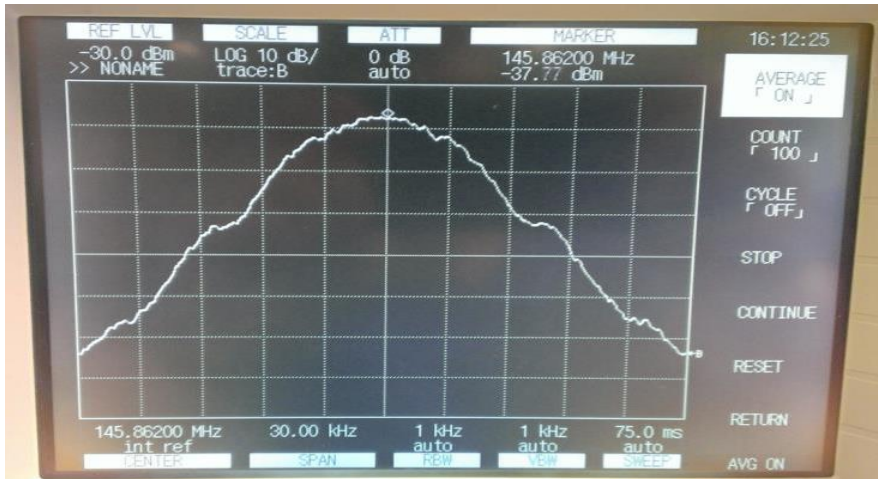
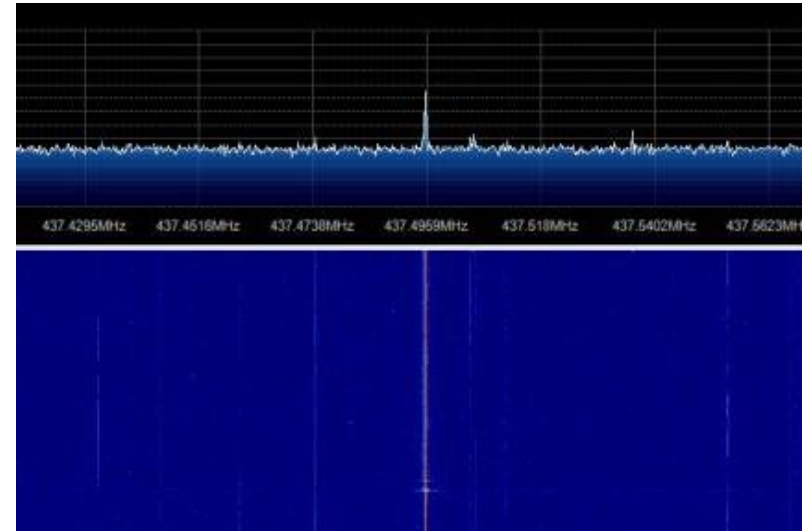
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- Routes the telecommands to the subsystems

Quick Interlude



STRaND-1: First Pass
No signal!

Quick Interlude



Introduction

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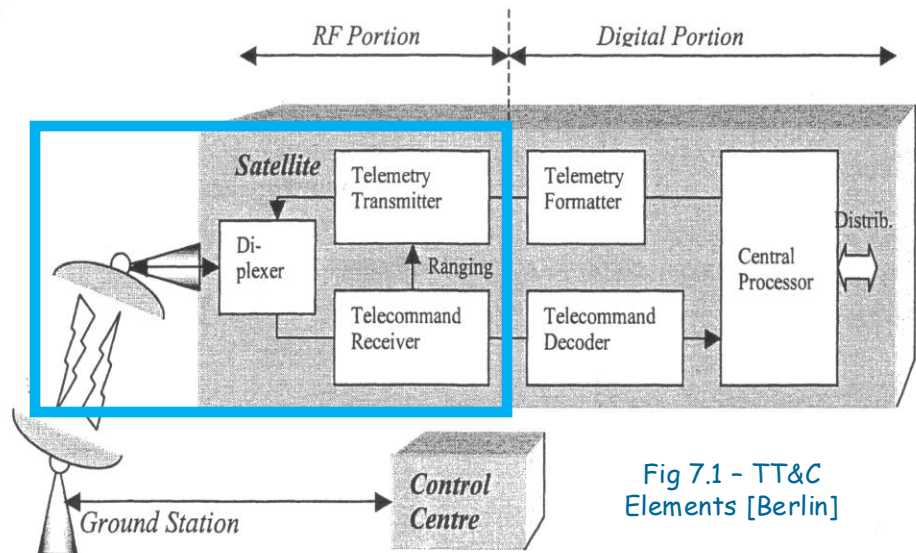


Fig 7.1 - TT&C Elements [Berlin]

TT&C made up of a Digital and a Radio Frequency (RF) portion.

Introduction

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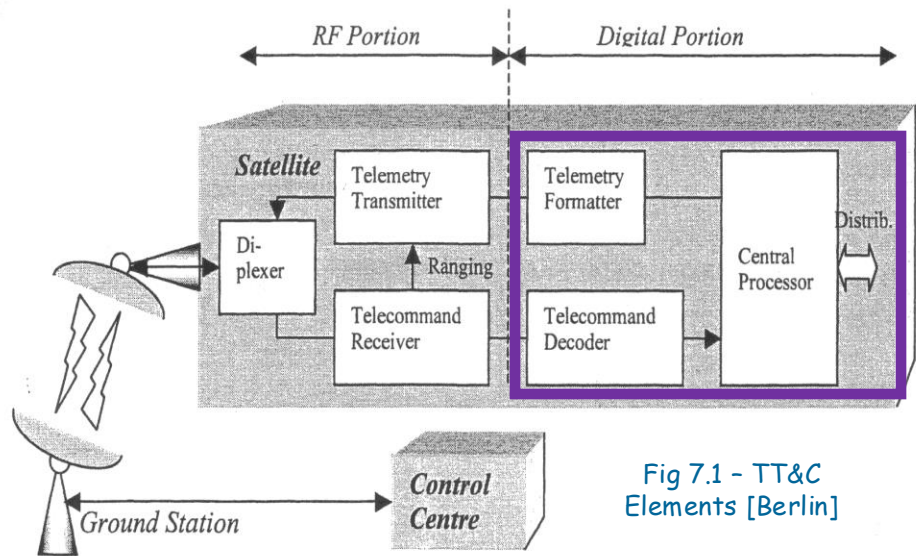
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TT&C made up of a Digital and a Radio Frequency (RF) portion.

Telemetry

- **Definition:**

- From “tele”=distance and “metry”= to measure hence, “measuring from a distance”
- In general, it refers to signals coming from the space platform.
- To mark a distinction between payload and bus data, the platform/bus data is also called “housekeeping telemetry” and can include:
 - Temperatures, on/off state of subsystems, voltages, currents, timing information...

Example Propulsion Controller Telemetry (TM)

Node Info		Channel Identifiers		Channel Format		Data Type		Calibration Equation			Named Values		Alarm Monitoring				Comment
This info must match the mission level naming convention Enter name of CAN node (usually module name) in accordance with mission naming convention. Text up to 15 characters. CAN node address from mission node address list. Enter decimal value		Display name for the tlm channel. Text up to 15 characters CAN Channel number. Enter decimal value		Use when 32 bit CAN channel is used for more than one tlm channel Identify the bit in the 32 bit channel at the start of the tlm required (0 - 31) Enter the length, in number of bits, of the channel (1 - 31)		Select from the drop down list, or enter value from the instruction sheet, to match raw data Select units from drop down list or enter value from instruction sheet. If no units required select "[]"		Equation format is $tlm\ value = ax^2 + bx + c$ for uncorrected counts enter $a=0, b=1, c=0$ Enter squared term in this column Enter linear multiplier in this column Enter constant term in this column			Use these columns when specific tlm values are identified by a name (eg if 1 = "On" and 0 = "Off") Specific tlm value (enter decimal value) Display name for this value enter text up to 15 characters		Use these columns to identify tlm channels which must remain within set limits. THINK CAREFULLY ABOUT THE VALUES YOU ENTER. The warning values will change the colour of the tlm display to show that the value is approaching its limit. The alarm limits will trigger alarms and may result in the s/c powering down certain modules. During flight operations this will result in the ground operators being paged by the ground station Enter decimal value for lower alarm limit Enter decimal value for lower warning limit (must be higher than Alarm Min) Enter decimal value for upper warning limit (must be lower than Alarm Max) Enter decimal value for upper alarm limit				Enter any comment that will be available using DB admin to explain telemetry point, or any important information regarding telemetry
NodeName	NodeNumber	TlmName	ChannelNumber	Starting bit	Length	Data Type	Units	Square	Linear	Constant	Value	ValName	AlarmMin	WarningMin	WarningMax	AlarmMax	Comment
Prop 0	32	Tank Pressure PT0	0	0	12	Unsigned Value	bar	0	0.1157	-6.9767			0	0	0	0	
Prop 0	32	Plenum Pressure PT2	1	0	12	Unsigned Value	bar	0	0.006	-0.4205			0	0	0	0	
Prop 0	32	Upstream Valve Temperature TS0	2	0	12	Word (16 bit)	°C	0	0.4	-273			0	0	0	0	
Prop 0	32	Plenum Temperature TS2	3	0	12	Word (16 bit)	°C	0	0.4	-273			0	0	0	0	
Prop 0	32	Plenum Pressure HMON	4	0	12	Unsigned Value	bar	0	0.006	-0.4205			0	0	0	0	
Prop 0	32	Plenum Pressure LMON	5	0	12	Unsigned Value	bar	0	0.006	-0.4205			0	0	0	0	
Prop 0	32	Current Monitor	6	0	12	Unsigned Value	mA	0	2.5573	14.862			0	0	0	0	
Prop 0	32	Thruster Temperature	7	0	12	Unsigned Value	°C	0	0.6001	-3.4289			0	0	0	0	
Prop 0	32	Unsolicited Telemetry ID Value	21	0	12	Unsigned Value	[]	0	1	0			0	0	0	0	
Prop 0	32	Mark Ratio	22	0	12	Unsigned Value	ms	0	10	0			0	0	0	0	
Prop 0	32	Space Ratio	23	0	12	Unsigned Value	ms	0	10	0			0	0	0	0	
Prop 0	32	Burn Time	24	0	12	Unsigned Value	ms	0	100	0			0	0	0	0	
Prop 0	32	Firmware Version No	25	0	12	Unsigned Value	[]	0	1	0			0	0	0	0	

Telecommands

- Definition
 - Frame of data containing command “words” for the execution of specific tasks onboard the spacecraft: attitude, operation...
 - Similarly to telemetry, dynamic frames are built to transmit the data
 - 3 types of commands are transmitted:
 - Low-level: electronic switches
 - High-level: electro-mechanical relays
 - Proportional: memory and computer loads, time-tag instructions

Example Propulsion Controller Telecommands (TC)

Node Info		Channel Identifiers		Default	Limits		Warning	Named Values			Comment
This info must match the mission level naming convention					Use these columns to set the range of possible input values for the command. Command software will not allow values outside of the specified limits to be sent to the s/c		Sends warning message to operator before actioning command	Use these columns when specific command values are identified by a name (eg if 1 = "On" and 0 = "Off")			
Enter name of CAN node (usually module name) in accordance with mission naming convention. Text up to 15 characters.	CAN node address from mission node address list. Enter decimal value	Display name for the command channel. Text up to 15 characters	CAN command number. Enter decimal value	The value here will be displayed as the default option in s/c command software. Enter decimal value if required	Enter decimal value for the minimum input	Enter decimal value for the maximum input	Select Yes or No. If Yes is selected a warning message will be displayed to the operator before sending the command	Entering yes in this column will restrict possible command entries to the named values only	Specific command value (enter decimal value)	Display name for this value enter text up to 15 characters	Enter any comment that will be available using DB admin to explain telecommand function, or any important information regarding telecommand
NodeName	NodeNumber	ChanName	ChannelNumber	DefaultValue	LimitMin	LimitMax	Warning	Values only	Value	ValName	Comment
Prop 0	32	Propellant type	0	0	0	1	No	Yes		0 Xenon	
Prop 0	32	Propellant type	0	0	0	1	No	Yes		1 Nitrogen	
Prop 0	32	Min Plenum Pressure	1	0	0	65536	No	Yes			
Prop 0	32	Max Plenum Pressure	2	0	0	65536	No	Yes			
Prop 0	32	Sol Operate Time	3	10	0	65536	No	Yes			
Prop 0	32	Low Pressure Warning	4	1	0	1	No	Yes		0 Disabled	
Prop 0	32	Low Pressure Warning	4	1	0	1	No	Yes		1 Enabled	
Prop 0	32	High Pressure Warning	5	1	0	65536	No	Yes		0 Disabled	
Prop 0	32	High Pressure Warning	5	1	0	65536	No	Yes		1 Enabled	
Prop 0	32	Operate Thruster (Sol 2 Reg)	6	0	0	65536	No	Yes			
Prop 0	32	Operate Thruster (Sol 0 Reg)	7	0	0	65536	No	Yes			
Prop 0	32	Thruster Heater 0	8	0	0	65536	No	Yes			
Prop 0	32	Burn Time	14	0	0	65536	No	Yes			
Prop 0	32	Operate Thruster m/s ratio (Sol2 Reg)	15	0	0	65536	No	Yes			
Prop 0	32	Operate Thruster m/s ratio (Sol0 Reg)	16	0	0	65536	No	Yes			
Prop 0	32	Operate Sol Valve 0	17	0	0	65536	No	Yes			
Prop 0	32	Operate Sol Valve 2	18	0	0	65536	No	Yes			

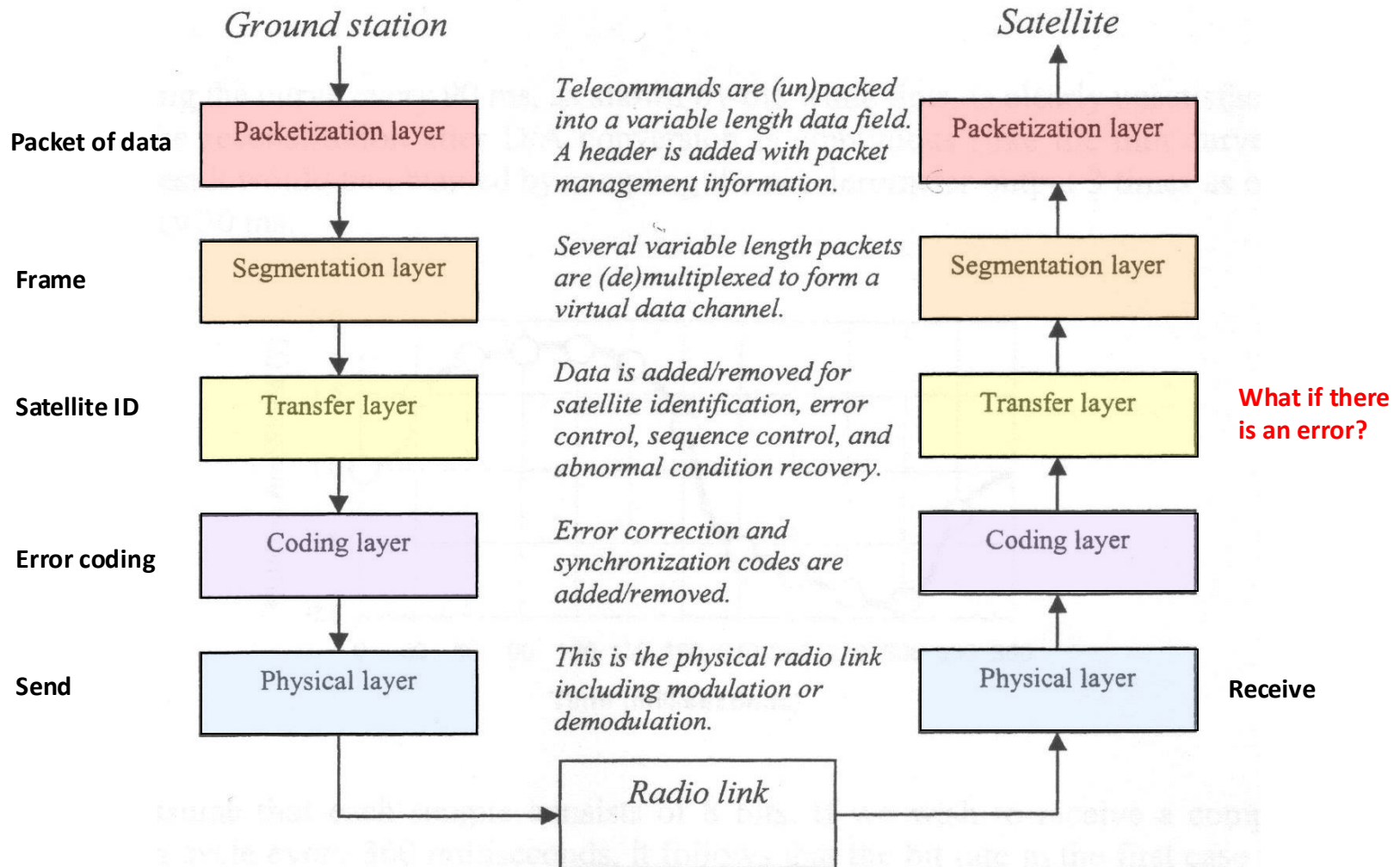
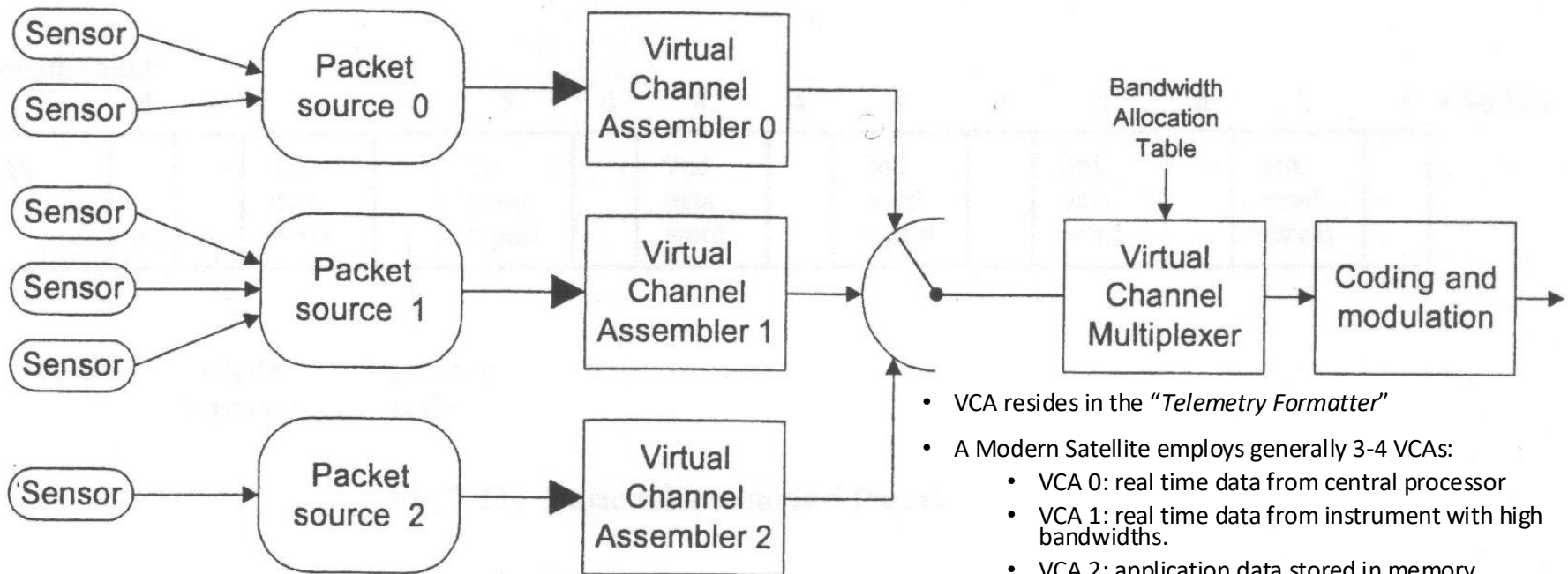


Fig 7.12 - Modern packet telecommand Protocol [Berlin]

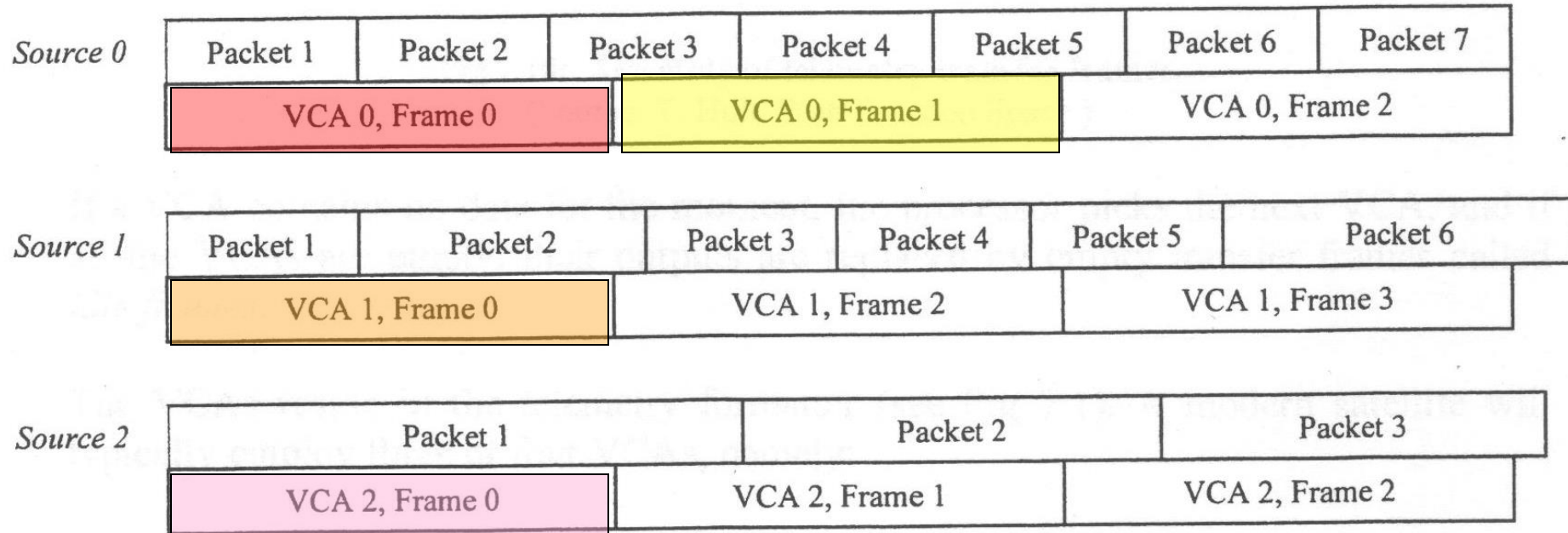
TMTC - Packet of Data & Frame

- Dynamically allocate the packet in the transmission process
 - *Virtual Channel Assemblers (VCA)* collect packets from various sources then generates a *"Telemetry Transfer Frame"*
 - A *Virtual Channel Multiplexer* decides which VCA to queue depending on the *Bandwidth Allocation Table (BAT)*



- VCA resides in the *"Telemetry Formatter"*
- A Modern Satellite employs generally 3-4 VCAs:
 - VCA 0: real time data from central processor
 - VCA 1: real time data from instrument with high bandwidths.
 - VCA 2: application data stored in memory
 - VCA 3: housekeeping data stored in memory
- If a VCA has no data, the next VCA is chosen. If all are empty, they are replaced by empty frames called *"idle frames"*

TMTC - Packet of Data & Frame



if the BAT stipulates 0, 1, 0, 2, ..., may yield the following TDM data stream:

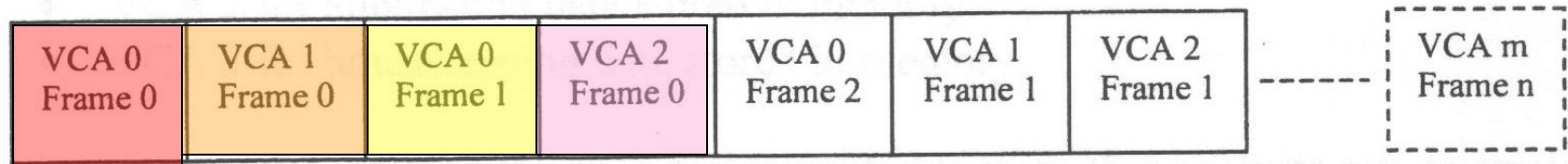


Fig 7.9 - Modern packet telemetry protocol [Berlin;Saab Ericsson Space]

TMTC - Error Detection and Correction



TMTC - Error Detection and Correction



TMTC - Modulation

AMPLITUDE MODULATION - ILLUSTRATION



TMTC - Encoding Digital Data



Noise and Losses

Noise?

WHAT IS NOISE IN
RADIO COMMUNICATIONS AND
WHO MAKES IT?



Noise and Losses



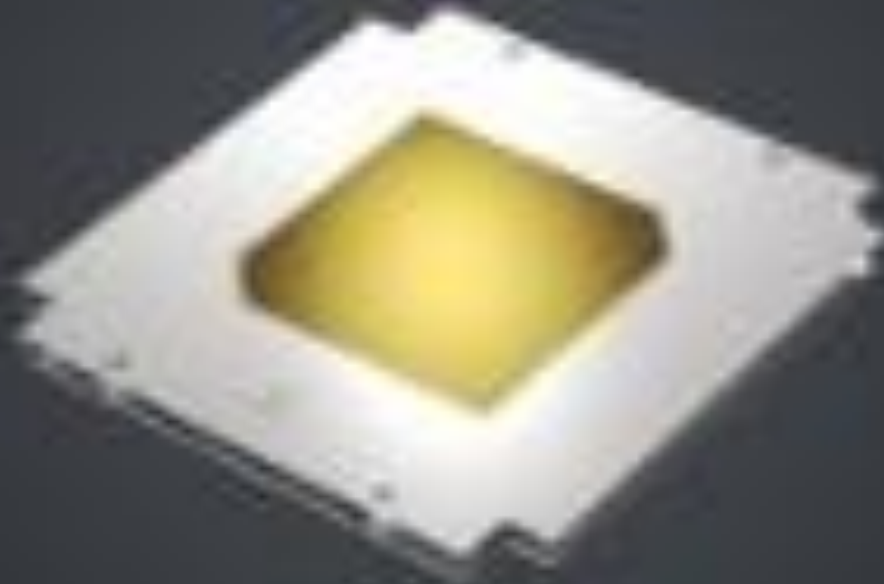
Emitting and Receiving Electromagnetic Waves

ANTENNAS — THE GENERAL IDEA



Emitting and Receiving Electromagnetic Waves

PATCH ANTENNAS



Link Budget

WHAT DOES THE RECEIVING ANTENNA SEE ?

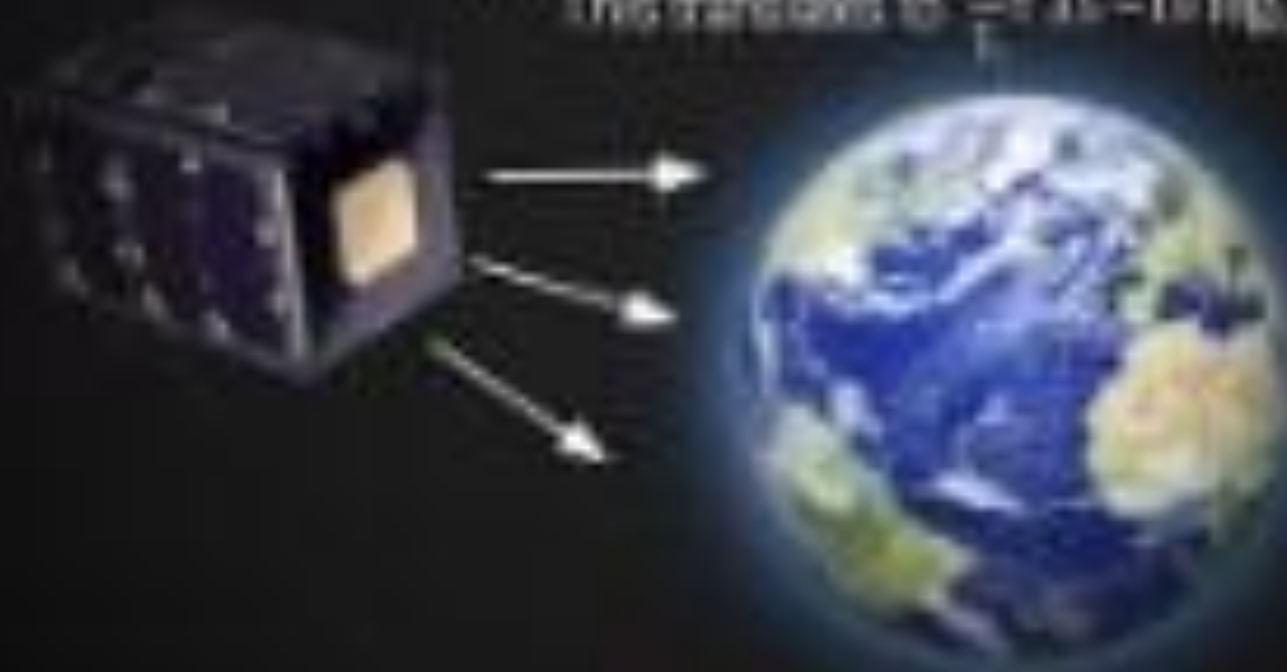


$$P_r = \frac{\text{EIRP}}{4\pi r^2} A_e = \frac{P_t G_t}{4\pi r^2} A_e$$

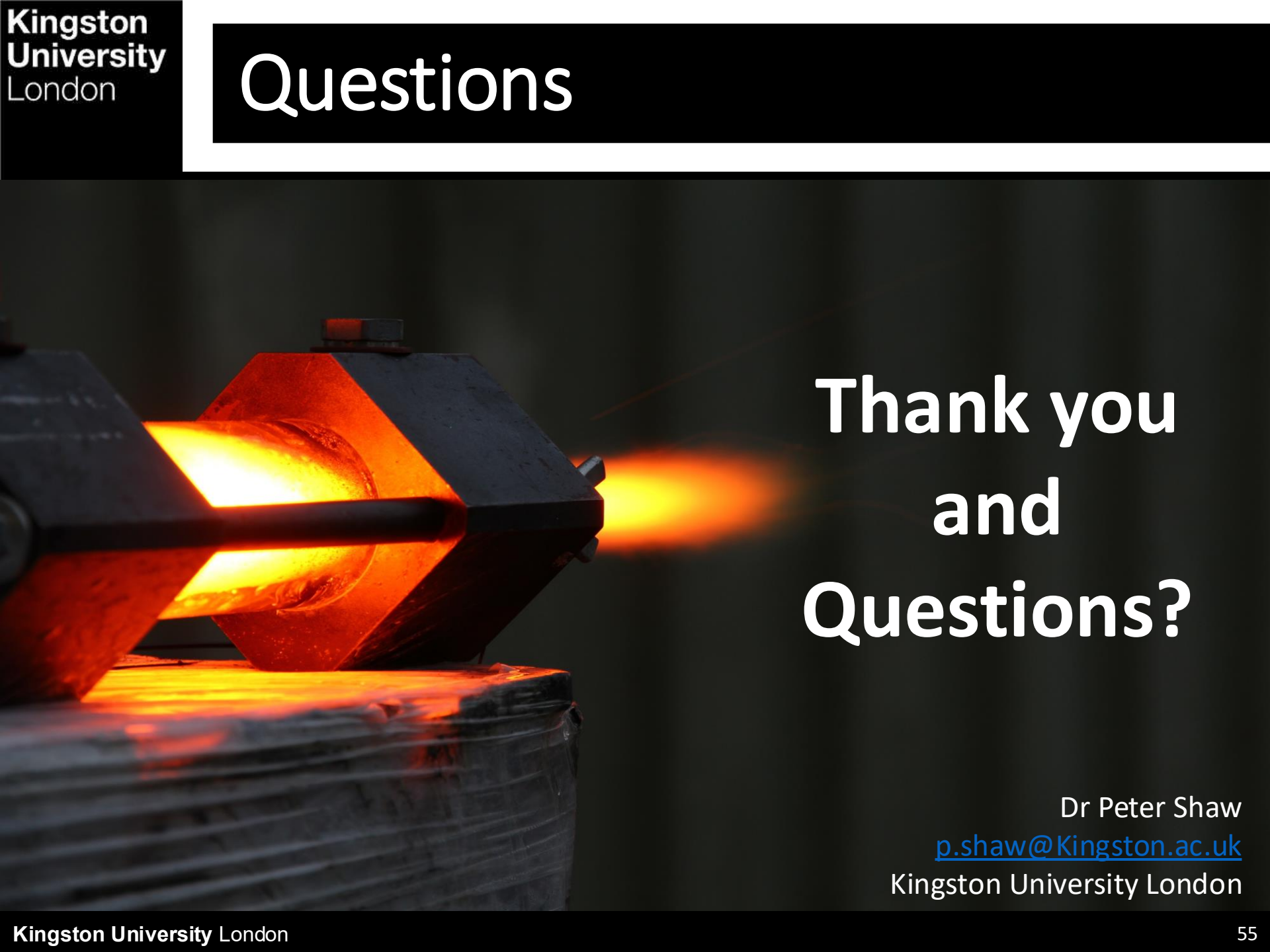
Link Budget

GEO EXAMPLE REVISITED

Consider the same GEO satellite and an equivalent system temperature of the whole chain of $T_e = 300$ K.
This translates to $\frac{S}{T} = 15 - 10 \log_{10} 300 = 18.2$ dB



Questions



Thank you
and
Questions?

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